MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE AGD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplexer module

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.47 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be AGD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/140-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier)

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6 2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements specified in 3.10.1 through 3.10.3.

3.10.1 Test patterns. A set of unique input patterns shall be generated capable of detecting all nonredundant failures in the module logic network. The test patterns shall be generated by assuming "stuck at H state" and "stuck at L state" failures in each of the basic logic functions and applying input patterns to detect these failures. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the tests of tables II and XXI satisfy or exceed these unique test patterns no additional tests shall be added, otherwise, the necessary tests shall be added to table XXI. "Stuck at H state" indicates a failure resulting in a steady high-level output. "Stuck at L state" indicates a failure resulting in a steady low-level output. Prior to any change in module design, which would necessitate a different set of test patterns, the vendor shall be required to submit to the procuring activity and the SEMP Quality Assurance Activity the proposed new set of test patterns for approval.

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3.10.2 Input requirements. The following input voltage levels shall be used for test patterns.

Low level - Input voltage shall be 0.8 V ±1 percent.
High level - Input voltage shall be 2.0 V ±1 percent.

3 10.3 Output requirements. The output requirements for the test patterns shall be as follows

Low level - Output voltage shall be 0 V minimum, 0.4 V maximum.
High level - Output voltage shall be 2.4 V minimum, V_{CC} maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4 4 Test procedures. The module shall be tested as specified in table I

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4 4 2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, C, or K are part of a sequence of tests, the order of testing shall be as follows. A, B, C, K where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

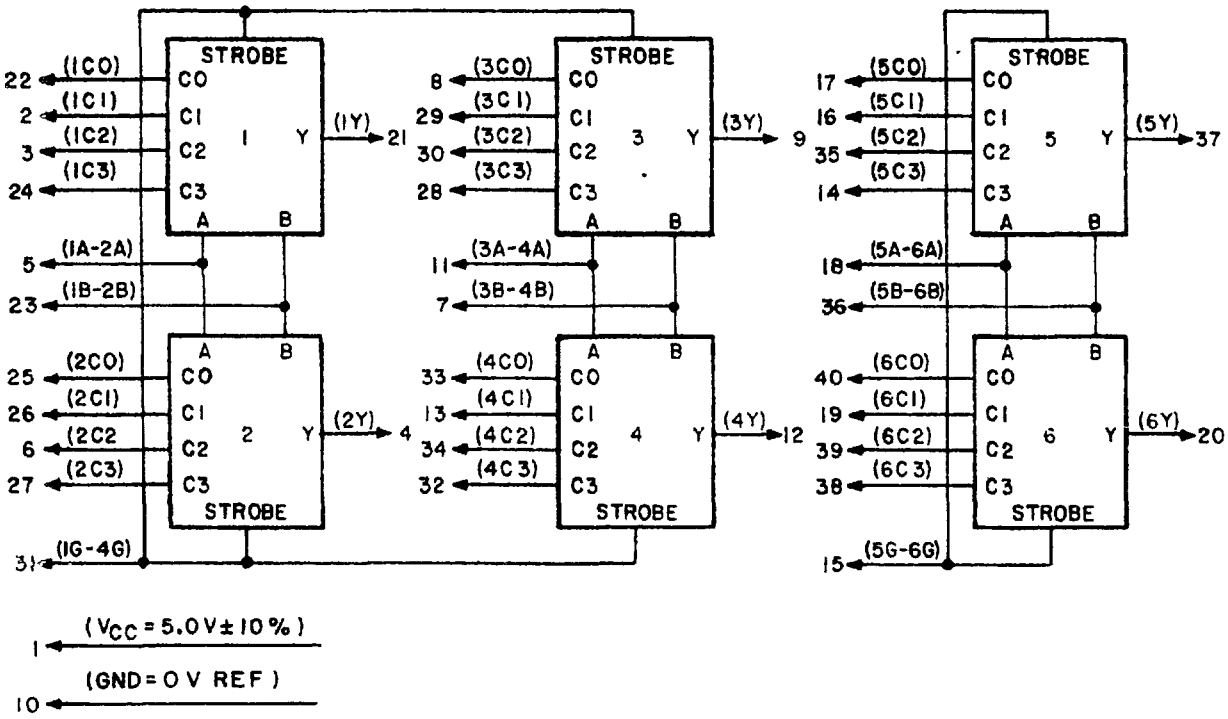
4 5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787 Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4 5.1 1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

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NOTES:
1. 1C0 - 6C0, 1C1 - 6C1, 1C2 - 6C2, 1C3 - 6C3 = Data inputs.
2. 1A - 2A, 3A - 4A, 5A - 6A, 1B - 2B, 3B - 4B, 5B - 6B = Common data select.
3. 1G - 4G, 5G - 6G = Common strobe inputs.
4. 1Y - 6Y = Multiplexer outputs.

FIGURE 1. AGD module function diagram, multiplexer.

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* TABLE I Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A6	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	mV
B1-B30	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
B31	Input breakdown current	IV	S	0	4	0	4	0	4	0	4	mA
B32	Input breakdown current	IV	S	0	2	0	2	0	2	0	2	mA
C	Isolation	V	S	10	-	1	-	10	-	1	∞	MΩ
D1-D30	Input capacitance	VI	Q	2	15	2	15	2	15	2	15	pF
D31	Input capacitance	VI	Q	2	60	2	60	2	60	2	60	pF
D32	Input capacitance	VI	Q	2	30	2	30	2	30	2	30	pF
E1-E32	Input clamp diode voltage	VII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
F1-F6	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1-G6	Output voltage, low level	VIII	X	0	400	0	400	0	400	0	400	mV
H1-H30	Input current, high level	IX	X	0	30	0	30	0	40	0	40	µA
H31	Input current, high level	IX	X	0	120	0	120	0	160	0	160	µA
H32	Input current, high level	IX	X	0	60	0	60	0	80	0	80	µA
J1-J30	Input current, low level	X	X	0.4	1.73	0.4	1.73	0.4	1.73	0.4	1.73	mA
J31	Input current, low level	X	X	1.6	6.92	1.6	6.92	1.6	6.92	1.6	6.92	mA
J32	Input current, low level	X	X	0.8	3.46	0.8	3.46	0.8	3.46	0.8	3.46	mA
K1-K6	Output short circuit current	XI	X	20	55	20	55	20	55	20	55	mA
L1	Power supply current, low level	XII	S	40	156	40	156	40	156	40	156	mA
L2	Power supply current, high level	XII	S	30	130	30	130	30	130	30	130	mA
M1-M6	Transition time (t_{TLH})	XIII	S	4	45	4	45	4	50	4	50	ns
N1-N6	Transition time (t_{THL})	XIV	S	4	20	4	20	4	25	4	25	ns
P1-P6	Propagation delay time (t_{PHL})	XV	S	7	33	7	33	7	35	7	35	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-37	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P7-P24	Propagation delay time (t_{PHL})	XV	Q	7	33	7	33	7	35	7	35	ns
P25-P30	Propagation delay time (t_{PHL})	XV	S	5	30	5	30	5	33	5	33	ns
P31-P48	Propagation delay time (t_{PHL})	XV	Q	5	30	5	30	5	33	5	33	ns
P49-P54	Propagation delay time (t_{PHL})	XV	S	10	41	10	41	10	44	10	44	ns
P55-P60	Propagation delay time (t_{PHL})	XV	Q	10	41	10	41	10	44	10	44	ns
P61-P66	Propagation delay time (t_{PHL})	XV	S	10	41	10	41	10	44	10	44	ns
P67-P72	Propagation delay time (t_{PHL})	XV	Q	10	41	10	41	10	44	10	44	ns
P73-P78	Propagation delay time (t_{PHL})	XV	S	7	41	7	41	7	44	7	44	ns
P79-P84	Propagation delay time (t_{PHL})	XV	Q	7	41	7	41	7	44	7	44	ns
P85-P90	Propagation delay time (t_{PHL})	XV	S	7	41	7	41	7	44	7	44	ns
P91-P96	Propagation delay time (t_{PHL})	XV	Q	7	41	7	41	7	44	7	44	ns
Q1-Q6	Propagation delay time (t_{PLH})	XVI	S	10	43	10	43	10	45	10	45	ns
Q7-Q24	Propagation delay time (t_{PLH})	XVI	Q	10	43	10	43	10	45	10	45	ns
Q25-Q30	Propagation delay time (t_{PLH})	XVI	S	5	29	5	29	5	31	5	31	ns
Q31-Q48	Propagation delay time (t_{PLH})	XVI	Q	5	29	5	29	5	31	5	31	ns
Q49-Q54	Propagation delay time (t_{PLH})	XVI	S	10	41	10	41	10	46	10	46	ns
Q55-Q60	Propagation delay time (t_{PLH})	XVI	Q	10	41	10	41	10	46	10	46	ns
Q61-Q66	Propagation delay time (t_{PLH})	XVI	S	10	41	10	41	10	46	10	46	ns
Q67-Q72	Propagation delay time (t_{PLH})	XVI	Q	10	41	10	41	10	46	10	46	ns
Q73-Q78	Propagation delay time (t_{PLH})	XVI	S	10	39	10	39	10	44	10	44	ns
Q79-Q84	Propagation delay time (t_{PLH})	XVI	Q	10	39	10	39	10	44	10	44	ns
Q85-Q90	Propagation delay time (t_{PLH})	XVI	S	10	39	10	39	10	44	10	44	ns
Q91-Q96	Propagation delay time (t_{PLH})	XVI	Q	10	39	10	39	10	44	10	44	ns

See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
1	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	mV
11-S6	Output crosstalk, low level	XVIII	Q	0	500	0	500	0	500	0	500	mV
7[illegible]-7C	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
2	Power supply filter capacitance	XX	S	2.2	6.6	2.2	6.6	2.2	6.6	2.2	6.6	µF
[illegible]-13	Functional test	XXI	X	← Test vectors must produce output vectors as specified in table XXI →								

1/ "A" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater)
at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/140-1 supersedes part number M28787/140. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage		5 Vdc ±10 percent
Maximum power dissipation		
Outputs at low level		858 mW (outputs not loaded)
Outputs at high level		715 mW (outputs not loaded)
Input voltage		
Low level		-0.5 V minimum, 0.8 V maximum
High level		2.0 V minimum, V_{CC} maximum
Input current		
A, B, and C inputs		
Low level	(V_{IL} = 0.0 V)	-1.73 mA maximum
	(V_{IL} = 0.4 V)	-1.60 mA maximum
High level	(V_{IH} = 2.4 V)	40 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Strobe input (1G-4G)		
Low level	(V_{IL} = 0.0 V)	-6.92 mA maximum
	(V_{IL} = 0.4 V)	-6.40 mA maximum
High level	(V_{IH} = 2.4 V)	160 µA maximum
	(V_{IH} = 5.5 V)	4.0 mA maximum
Strobe input (5G-6G)		
Low level	(V_{IL} = 0.0 V)	-3.46 mA maximum
	(V_{IL} = 0.4 V)	-3.20 mA maximum
High level	(V_{IH} = 2.4 V)	80 µA maximum
	(V_{IH} = 5.5 V)	2.0 mA maximum
Output voltage		
Low level	(I_{OL} = 16 mA)	0.4 V maximum
High level	(I_{OH} = -800 µA)	2.4 V minimum
Output drive current maximum capability		
Low level	(V_{OL} = 0.4 V)	16 mA
High level	(V_{OH} = 2.4 V)	-800 µA
Propagation delay time (t_{PLH})		
C_L = 100 pF		
Data to output		5.0 ns minimum, 31.0 ns maximum
Strobe to output		10.0 ns minimum, 45.0 ns maximum
Select to output (not inverted)		10.0 ns minimum, 46.0 ns maximum
Select to output (inverted)		10.0 ns minimum, 44.0 ns maximum

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Propagation delay time (t_{PHL})
C_L = 100 pF

Data to output	5.0 ns minimum, 33 ns maximum
Strobe to output	7.0 ns minimum, 35 ns maximum
Select to output (not inverted)	10.0 ns minimum, 44 ns maximum
Select to output (inverted)	7.0 ns minimum, 44 ns maximum

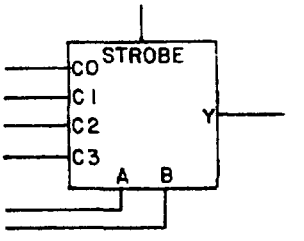
* 6.5 Module function. This module contains six 4-line to 1-line data multiplexer circuits which are paired to share common select lines. One strobe line is shared by four circuits and another strobe line is provided for the remaining two circuits. The logic diagram is shown on figure 1.

6.5.1 Input to output relationship. Each multiplexer circuit has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/-3/

Strobe	Select		Data inputs				Output
G	A	B	C0	C1	C2	C3	Y
H	X	X	X	X	X	X	L
L	L	L	L	X	X	X	L
L	L	L	H	X	X	X	H
L	H	L	X	L	X	X	L
L	H	L	X	H	X	X	H
L	L	H	X	X	L	X	L
L	L	H	X	X	H	X	H
L	H	H	X	X	X	L	L
L	H	H	X	X	X	H	H

1/ H = High level.
2/ L = Low level.
3/ X = Don't care.



6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

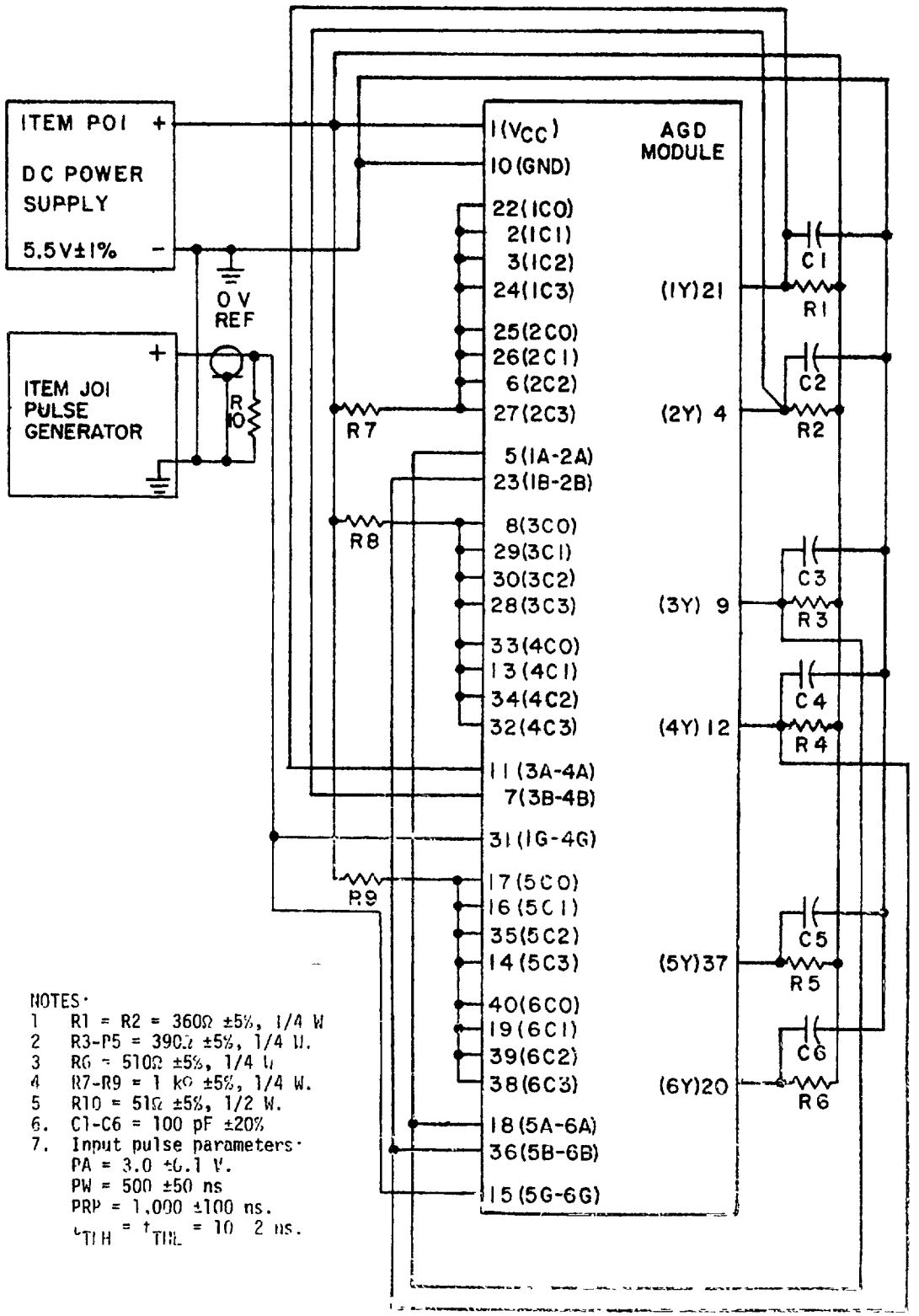
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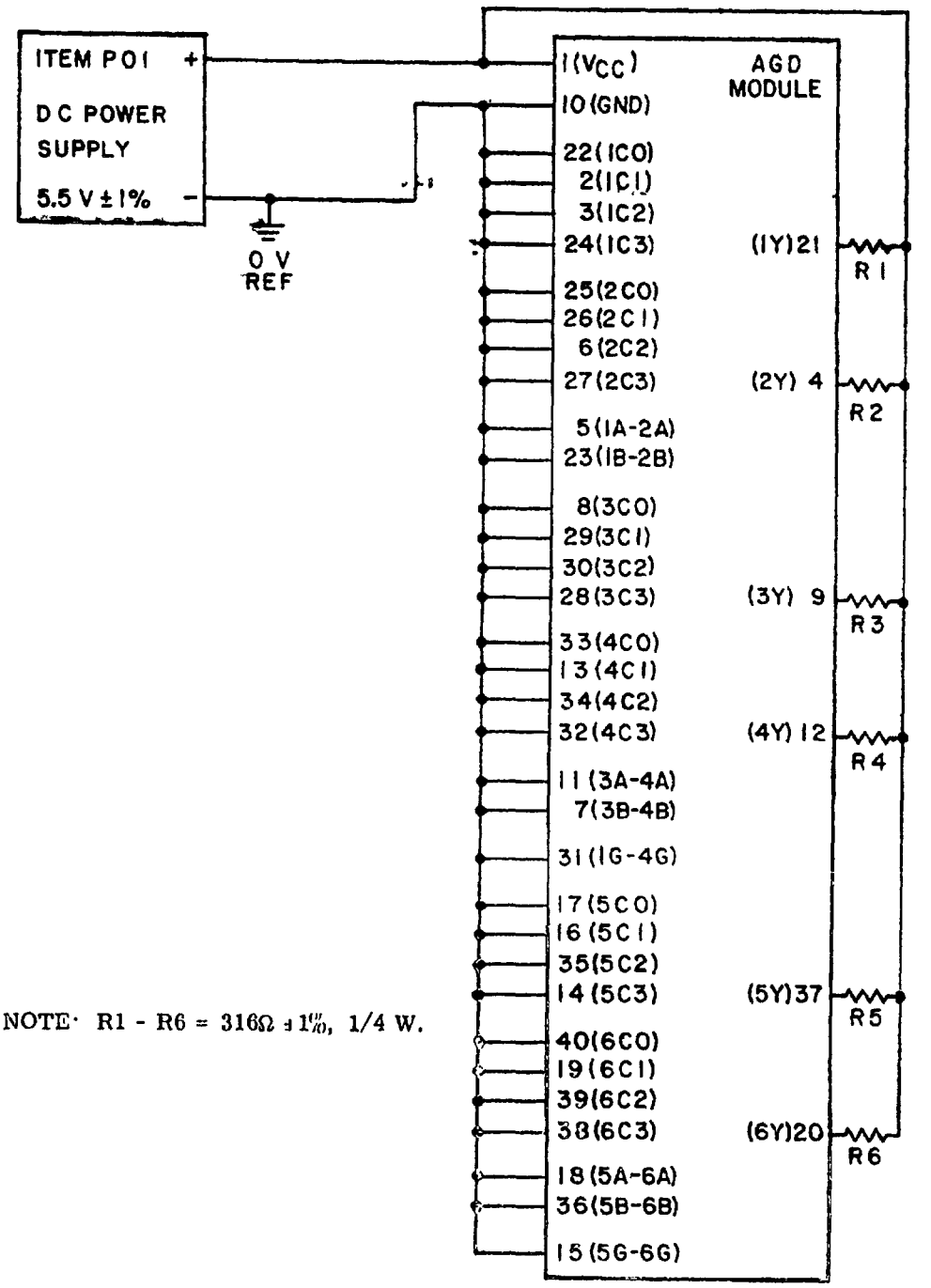
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NOTES:
1. R1 = R2 = 360Ω ±5%, 1/4 W
2. R3-R5 = 390Ω ±5%, 1/4 W.
3. R6 = 510Ω ±5%, 1/4 W
4. R7-R9 = 1 kΩ ±5%, 1/4 W.
5. R10 = 51Ω ±5%, 1/2 W.
6. C1-C6 = 100 pF ±20%
7. Input pulse parameters:
   PA = 3.0 ±0.1 V.
   PW = 500 ±50 ns
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.

FIGURE 2. Operating, temperature and life test circuit

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NOTE: R1 - R6 = 316Ω ± 1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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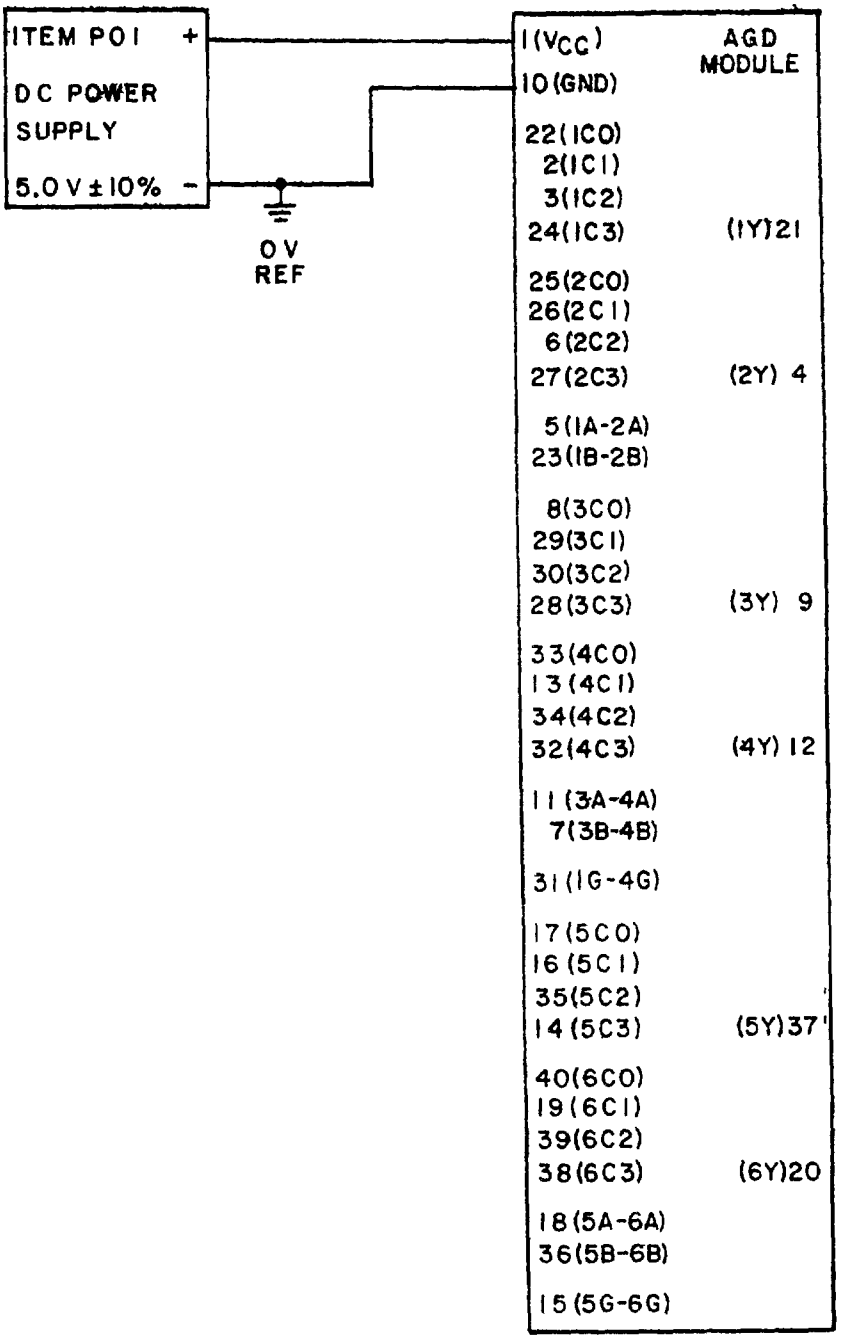


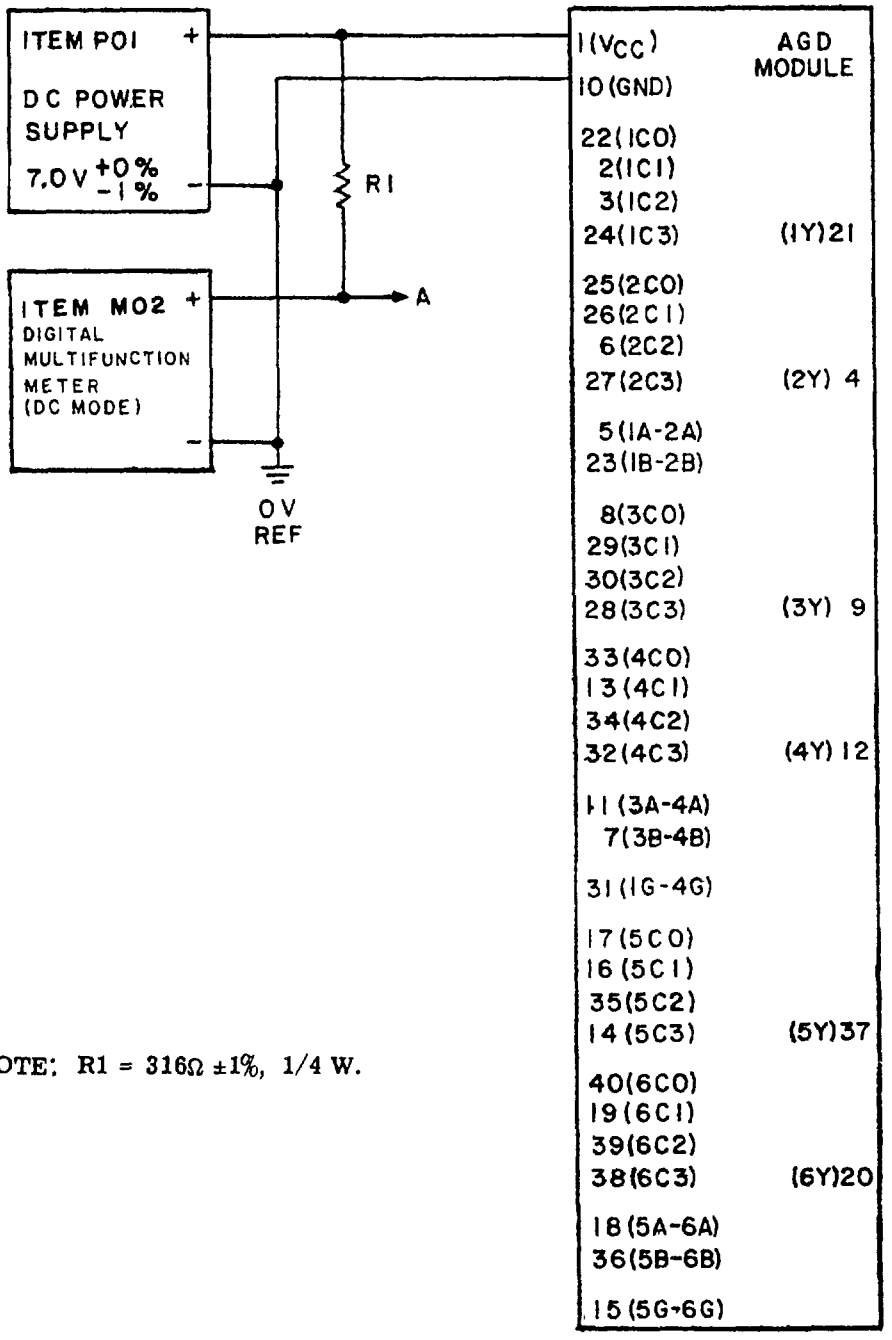
FIGURE 4. Vibration test circuit.

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* TABLE III. V_{OUT} (low-loade est.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1Y V_{OUT} (low loaded)	21	10	800	mV	5
A2	2Y	4	↓	↓	↓	↓
A3	3Y	9				
A4	4Y	12				
A5	5Y	37				
A6	6Y	20				

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NOTE: R1 = 316Ω ±1%, 1/4 W.

* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. *Input breakdown current test.*

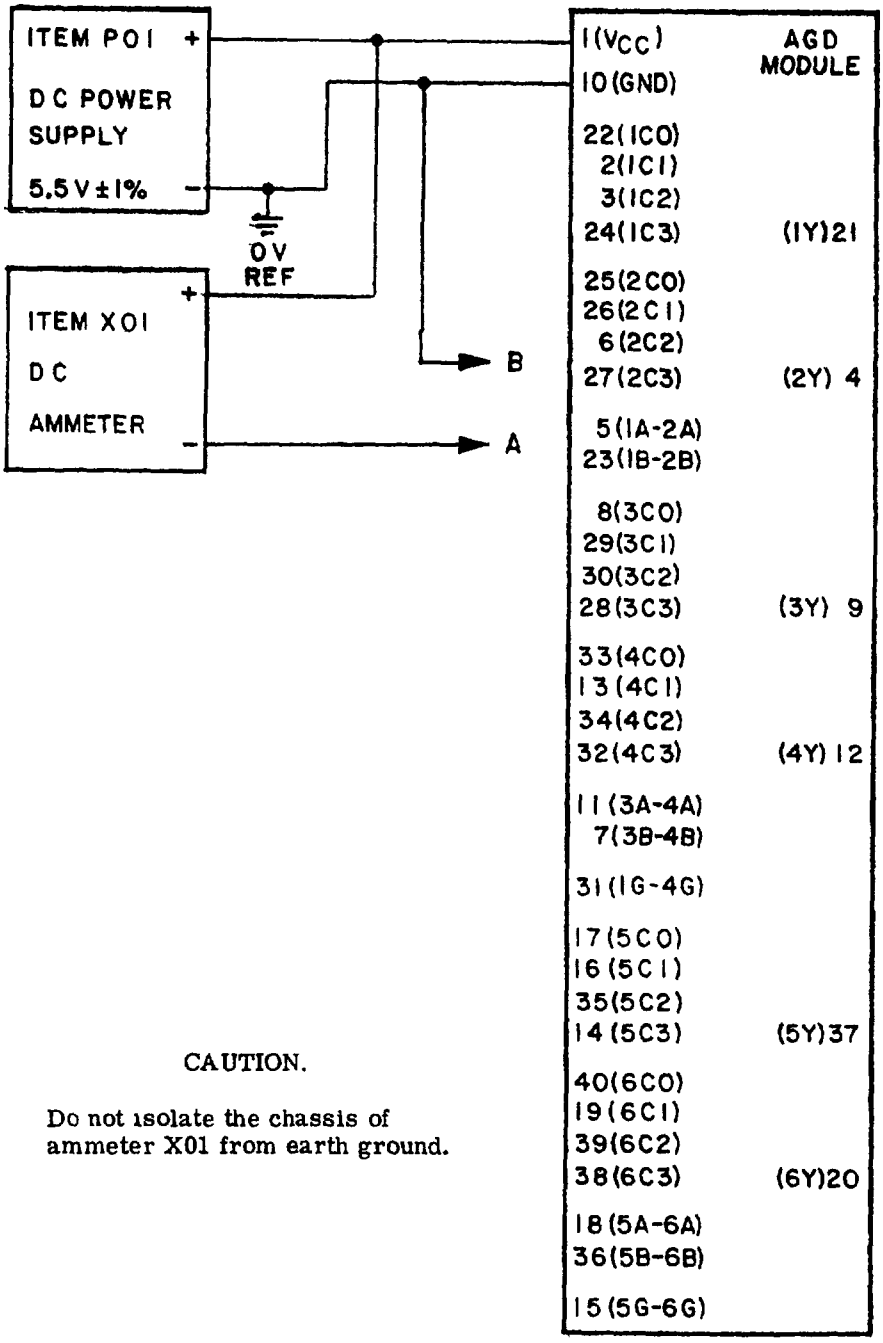
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1C0 Input breakdown current	22	Open	0	1	mA	6
B2	1C1	2	5				
B3	1C2	3	23				
B4	1C3	24	5,23				
B5	2C0	25	Open				
B6	2C1	26	5				
B7	2C2	6	23				
B8	2C3	27	5,23				
B9	3C0	8	Open				
B10	3C1	29	11				
B11	3C2	30	7				
B12	3C3	28	7,11				
B13	4C0	33	Open				
B14	4C1	13	11				
B15	4C2	34	7				
B16	4C3	32	7,11				
B17	5C0	17	Open				
B18	5C1	16	18				
B19	5C2	35	36				
B20	5C3	14	18,36				
B21	6C0	40	Open				
B22	6C1	19	18				
B23	6C2	39	36				
B24	6C3	38	18,36				

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B25	1A-2A Input breakdown current	5	Open	0	1	mA	6
B26	1B-2B	23					
B27	3A-4A	11					
B28	3B-4B	7					
B29	5A-6A	18					
B30	5B-6B	36					
B31	1G-4G	31			4		
B32	5G-6G	15			2		

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CAUTION.

Do not isolate the chassis of ammeter X01 from earth ground.

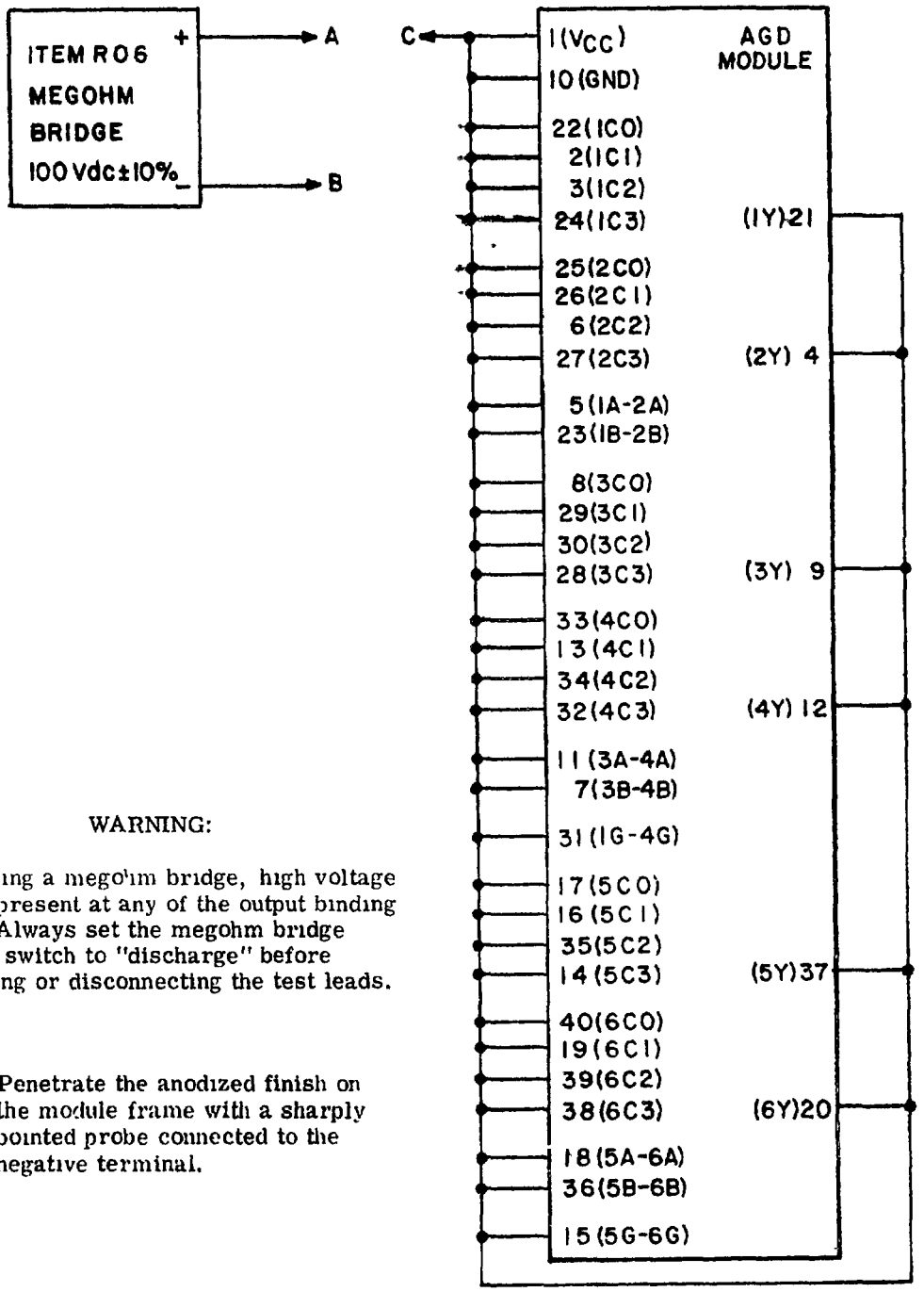
* FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation	C	Frame	10	∞	MΩ	7

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Input capacitance test.

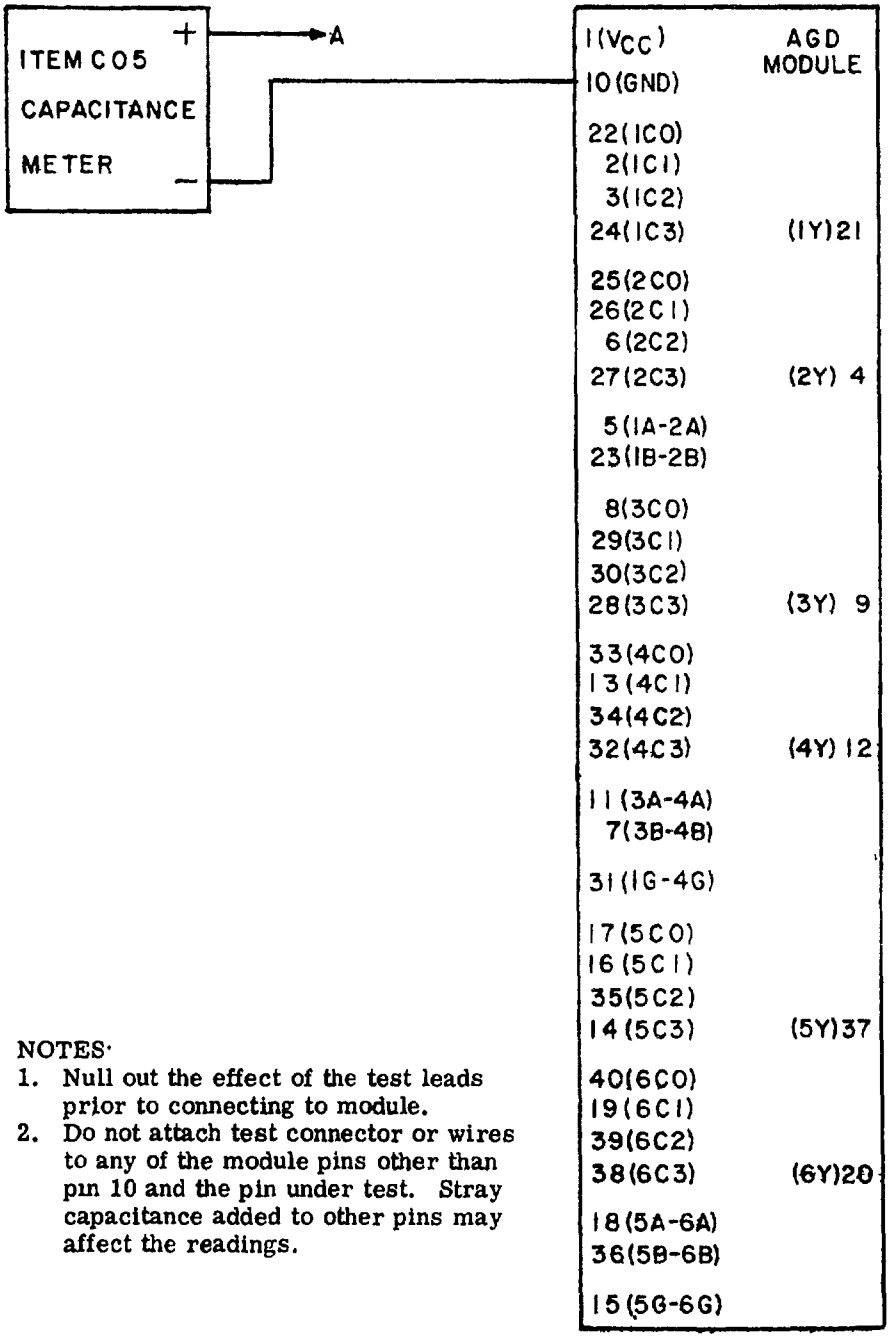
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	1C0 Input capacitance	22	2	15	pF	8
D2	1C1	2				
D3	1C2	3				
D4	1C3	24				
D5	2C0	25				
D6	2C1	26				
D7	2C2	6				
D8	2C3	27				
D9	3C0	8				
D10	3C1	29				
D11	3C2	30				
D12	3C3	28				
D13	4C0	33				
D14	4C1	13				
D15	4C2	34				
D16	4C3	32				
D17	5C0	17				
D18	5C1	16				
D19	5C2	35				
D20	5C3	14				
D21	6C0	40				
D22	6C1	19				
D23	6C2	39				
D24	6C3	38				

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TABLE VI. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D25	1A-2A Input capacitance	5	2	15	pF	8
D26	1B-2B	23				
D27	3A-4A	11				
D28	3B-4B	7				
D29	5A-6A	18				
D30	5B-6B	36				
D31	1G-4G	31		60		
D32	5G-6G	15		30		

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NOTES:
1. Null out the effect of the test leads prior to connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

* FIGURE 8. Input capacitance test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

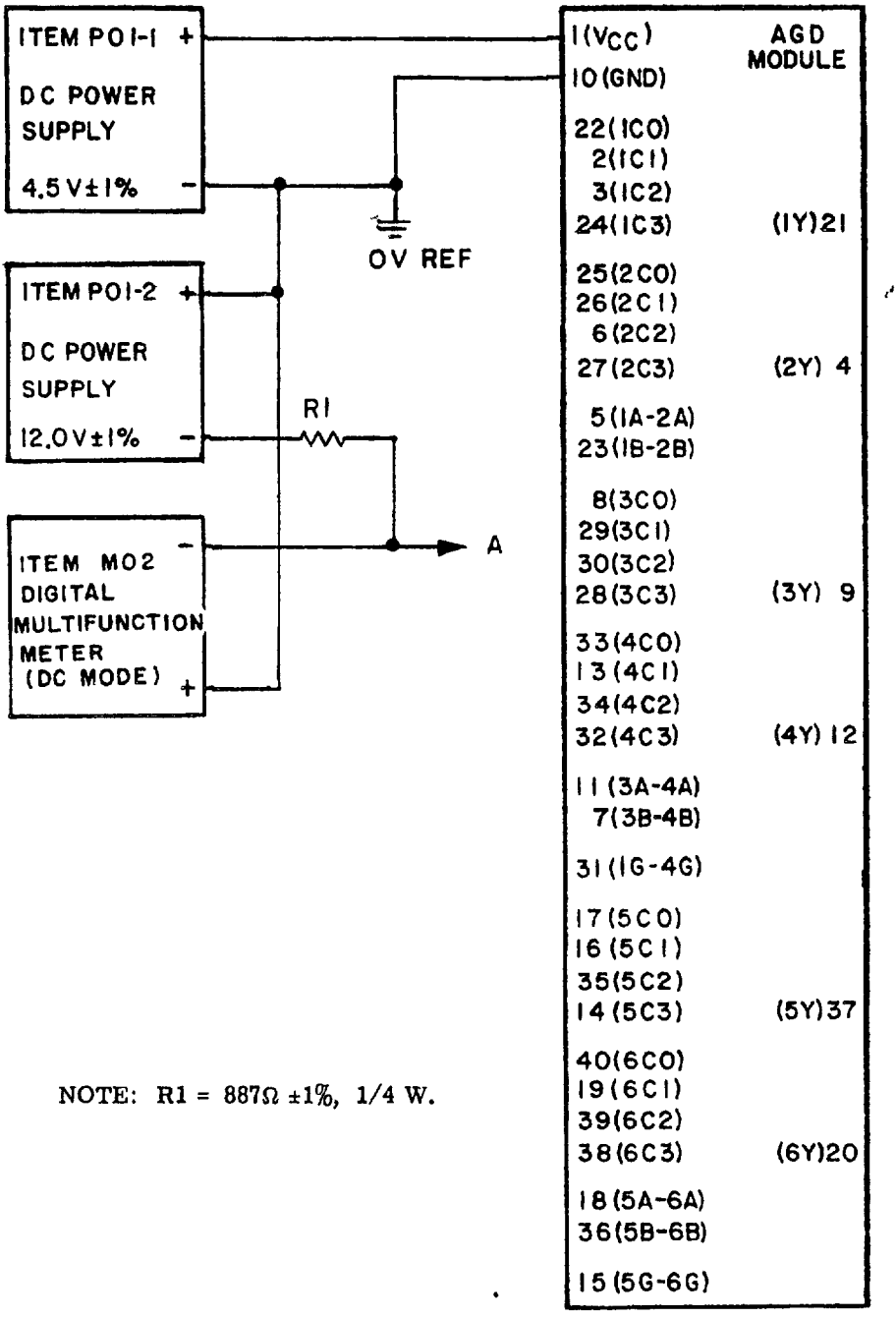
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	1C0 Input clamp diode voltage	22	-0.4	-1.5	V	9
E2	1C1	2				
E3	1C2	3				
E4	1C3	24				
E5	2C0	25				
E6	2C1	26				
E7	2C2	6				
E8	2C3	27				
E9	3C0	8				
E10	3C1	29				
E11	3C2	30				
E12	3C3	28				
E13	4C0	33				
E14	4C1	13				
E15	4C2	34				
E16	4C3	32				
E17	5C0	17				
E18	5C1	16				
E19	5C2	35				
E20	5C3	14				
E21	6C0	40				
E22	6C1	19				
E23	6C2	39				
E24	6C3	38				

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TABLE VII. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E25	1A-2A Input clamp diode voltage	5	-0.4	-1.5	V	9
E26	1B-2B	23				
E27	3A-4A	11				
E28	3B-4B	7				
E29	5A-5A	18				
E30	5B-6B	36				
E31	1G-4G	31				
E32	5G-6G	15				

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NOTE: R1 = 887Ω ±1%, 1/4 W.

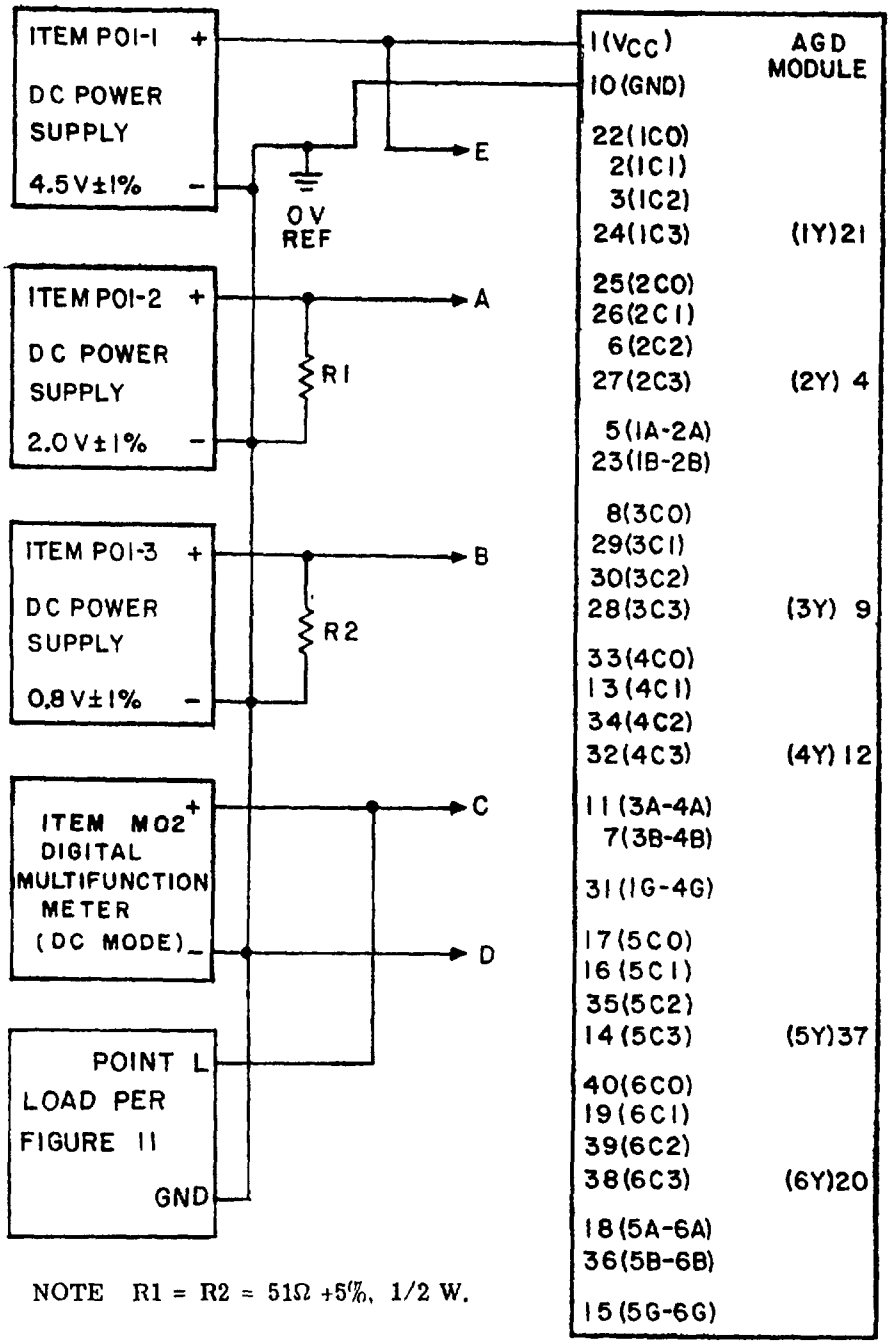
* FIGURE 9. Input clamp diode voltage test circuit for table VII.

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* TABLE VIII. Output voltage (high-level and low-level) test.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F1	1Y	Output voltage, high level	22	5,23, 31	21	2,3,24	Open	2.4	4.5	V	10
F2	2Y		25	5,23, 31	4	6,26, 27					
F3	3Y		8	7,11, 31	9	28,29, 30					
F4	4Y		33	7,11, 31	12	13,32, 34					
F5	5Y		17	15,18, 36	37	14,16, 35					
F6	6Y		40	15,18, 36	20	19,38, 39					
G1	1Y	Output voltage, low level	5,23	24	21	31	2,3,22	0	400	mV	
G2	2Y		5,23	27	4	31	6,25, 26				
G3	3Y		7,11	28	9	31	8,29, 30				
G4	4Y		7,11	32	12	31	13,33, 34				
G5	5Y		18,36	14	37	15	16,17, 35				
G6	6Y		18,36	38	20	15	19,39, 40				

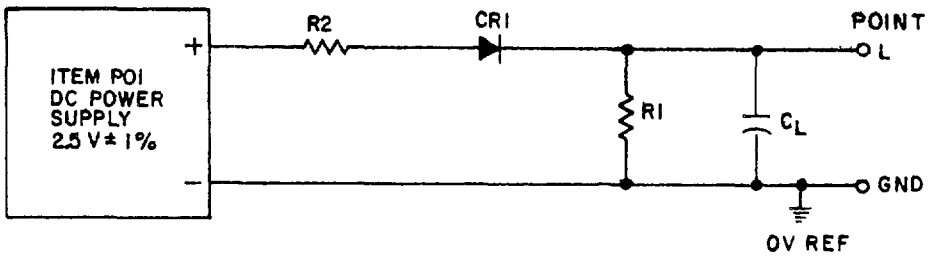
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NOTE R1 = R2 = 51Ω +5%, 1/2 W.

* FIGURE 10. Output voltage (high-level and low-level) test circuit for table VIII.

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NOTES:
1. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions M1-M6, N1-N6, P1-P96, and Q1-Q96.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions S1-S6 and T1-T6.
3. CR1 = 1N4150, or equivalent.
4. R1 = 3.01 kΩ ±1%, 1/8 W.
5. R2 = select value for a current flow of 16.0 mA ±5% from point "L" when "L" is held at +0.4 V.
6. C_L is not required for dc electrical test conditions F1-F6 and G1-G6.
7. The maximum lead length from point "L" to the appropriate module pin shall be 3 inches (76.2 mm).
8. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 11 Logic load circuit.

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1C0 Input current, high level	22	Open	Open	0	30	µA	12
H2	1C1	2		5				
H3	1C2	3		23				
H4	1C3	24		5,23				
H5	2C0	25		Open				
H6	2C1	26		5				
H7	2C2	6		23				
H8	2C3	27		5,23				
H9	3C0	8		Open				
H10	3C1	29		11				
H11	3C2	30		7				
H12	3C3	28		7,11				
H13	4C0	33		Open				
H14	4C1	13		11				
H15	4C2	34		7				
H16	4C3	32		7,11				
H17	5C0	17		Open				
H18	5C1	16		18				
H19	5C2	35		36				
H20	5C3	14		18,36				
H21	6C0	40		Open				
H22	6C1	19		18				
H23	6C2	39		36				
H24	6C3	38		18,36				

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TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H25	1A-2A Input current, high level	5	Open	Open	0	30	µA	12
H26	1B-2B	23						
H27	3A-4A	11						
H28	3B-4B	7						
H29	5A-6A	18						
H30	5B-6B	36						
H31	1G-4G	31				120		
H32	5G-6G	15				60		

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TABLE X. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1C0 Input current, low level	Open	22	5,23,31	0.4	1.73	mA	12
J2	1C1		2	23,31				
J3	1C2		3	5,31				
J4	1C3		24	31				
J5	2C0		25	5,23,31				
J6	2C1		26	23,31				
J7	2C2		6	5,31				
J8	2C3		27	31				
J9	3C0		8	7,11,31				
J10	3C1		29	7,31				
J11	3C2		30	11,31				
J12	3C3		28	31				
J13	4C0		33	7,11,31				
J14	4C1		13	7,31				
J15	4C2		34	11,31				
J16	4C3		32	31				
J17	5C0		17	15,18,36				
J18	5C1		16	15,36				
J19	5C2		35	15,18				
J20	5C3		14	15				
J21	6C0		40	15,18,36				
J22	6C1		19	15,36				
J23	6C2		39	15,18				
J24	6C3		38	15				

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TABLE X. Input current (low-level) test - Continued.

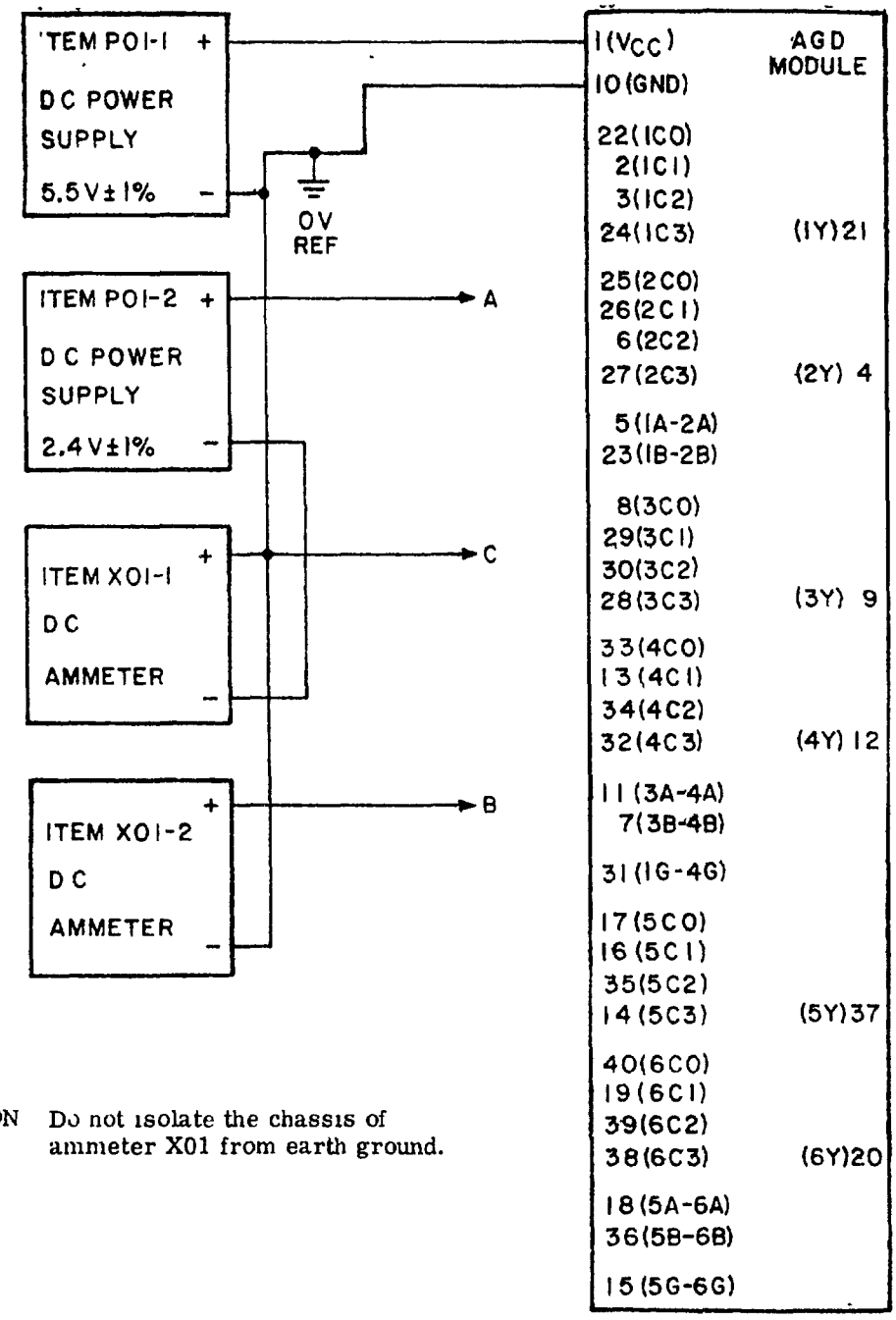
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J25	1A-2A Input current, low level	Open	5	Open	0.4	1.73	mA	12
J26	3A-4A		11					
J27	5A-6A		18					
J28	1B-2B		23					
J29	3B-4B		7					
J30	5B-6B		36					
J31	1G-4G		31		1.6	6.92		
J32	5G-6G		15		0.8	3.46		

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TABLE XI. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1Y Output short circuit current	Open	21	31	20	55	mA	12
K2	2Y		4					
K3	3Y		9					
K4	4Y		12					
K5	5Y		37	15				
K6	6Y		20	15				

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CAUTION Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required. Use item X01-2 for tables X and XI, use item X01-1 for table IX.
2. With point "A" disconnected, item X01-1 should indicate zero current on the 1 µA range.
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

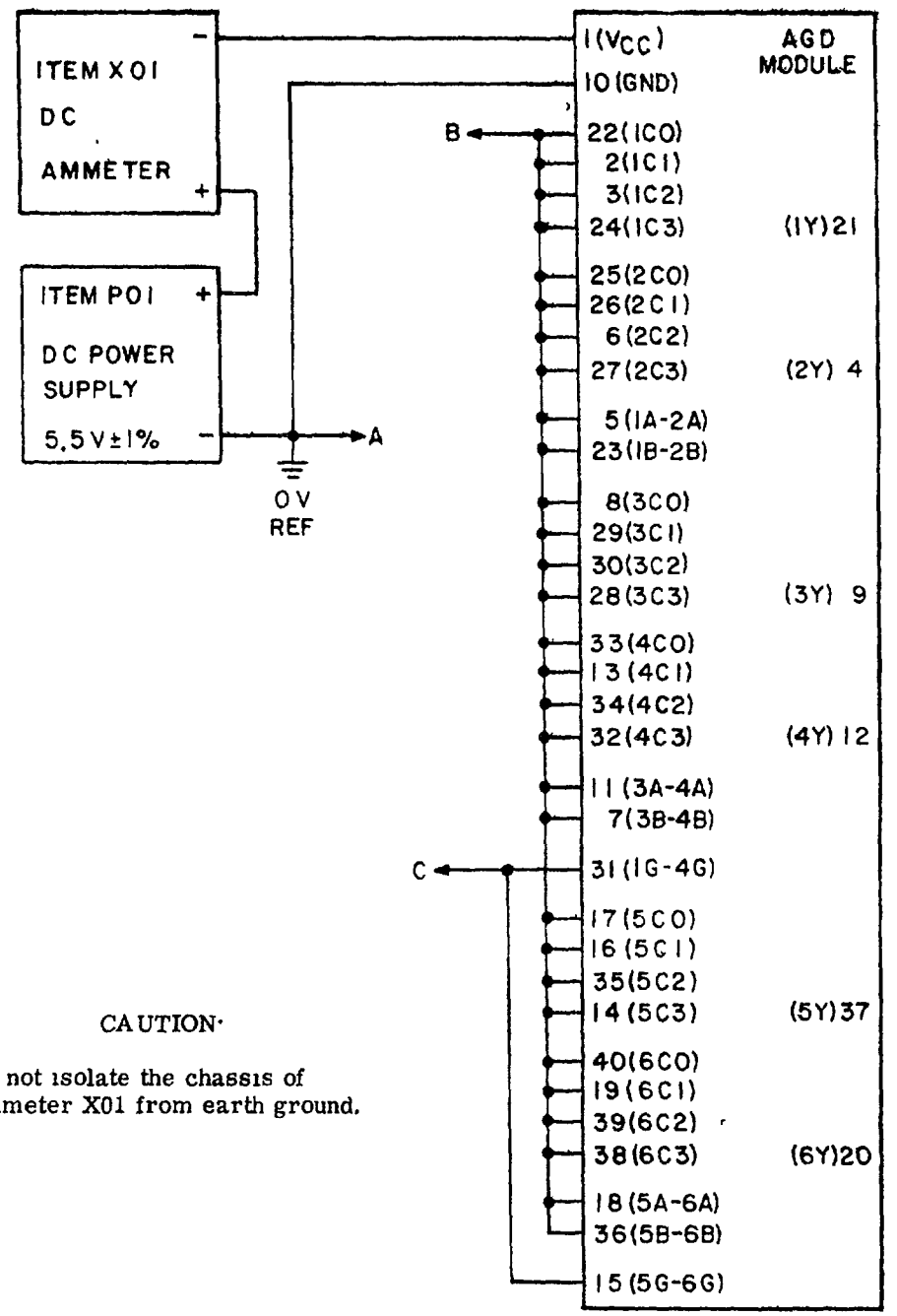
* FIGURE 12. Input current (high-level and low-level) and output short circuit current test circuit for tables IX, X, and XI.

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TABLE XII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Power supply current, low level	B,C	40	156	mA	13
L2	Power supply current, high level	C	30	130	mA	13

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

* FIGURE 13. Power supply current test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1Y Transition time (t_{TLH})	Open	Open	31	21	4	45	ns	14,15
M2	2Y	↓	↓	↓	4	↓	↓	↓	↓
M3	3Y	↓	↓	↓	9	↓	↓	↓	↓
M4	4Y	↓	↓	↓	12	↓	↓	↓	↓
M5	5Y	↓	↓	15	37	↓	↓	↓	↓
M6	6Y	↓	↓	15	20	↓	↓	↓	↓

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* TABLE XIV. Transition time (t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1Y Transition time (t_{THL})	Open	Open	31	21	4	20	ns	14,15
N2	2Y	↓	↓	↓	4	↓	↓	↓	↓
N3	3Y	↓	↓	↓	9	↓	↓	↓	↓
N4	4Y	↓	↓	↓	12	↓	↓	↓	↓
N5	5Y	↓	↓	15	37	↓	↓	↓	↓
N6	6Y	↓	↓	15	20	↓	↓	↓	↓

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1G-4G Propagation delay time to 1Y (t_{PHL})	22	5,23	31	21	7	33	ns	14,17
P2	1G-4G to 2Y	25	5,23		4				
P3	1G-4G to 3Y	8	7,11		9				
P4	1G-4G to 4Y	33	7,11		12				
P5	5G-6G to 5Y	17	18,36	15	37				
P6	5G-6G to 6Y	40	18,36	15	20				
P7	1G-4G to 1Y	2,5	23	21	21				
P8	1G-4G to 2Y	26,5	23		4				
P9	1G-4G to 3Y	29,11	7		9				
P10	1G-4G to 4Y	13,11	7		12				
P11	5G-6G to 5Y	16,10	36	15	37				
P12	5G-6G to 6Y	19,10	36	15	20				
P13	1G-4G to 1Y	3,23	5	31	21				
P14	1G-4G to 2Y	3,23	5	31	4				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P15	1G-4G to 3Y; Propagation delay time (t_{PHL})	30,7	11	31	9	7	33	ns	14,17
P16	1G-4G to 4Y	34,7	11	31	12				
P17	5G-6G to 5Y	35,36	18	15	37				
P18	5G-6G to 6Y	39,36	18	15	20				
P19	1G-4G to 1Y	5,24,23	Open	31	21				
P20	1G-4G to 2Y	5,27,23			4				
P21	1G-4G to 3Y	7,28,11			9				
P22	1G-4G to 4Y	7,32,11			12				
P23	5G-6G to 5Y	18,14,36		15	37				
P24	5G-6G to 6Y	18,38,36		15	20				
P25	1C0 to 1Y	Open	5,23,31	22	21	5	30		14,16
P26	2C0 to 2Y		5,23,31	25	4				
P27	3C0 to 3Y		7,11,31	8	9				
P28	4C0 to 4Y		7,11,31	33	12				
P29	5C0 to 5Y		18,15,36	17	37				
P30	6C0 to 6Y		18,15,36	40	20				
P31	1C1 to 1Y	5	31,23	2	21				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P32	2C1 to 2Y Propagation delay time (t_{PHL})	5	31,23	26	4	5	30	ns	14,16
P33	3C1 to 3Y	11	31,7	29	9				
P34	4C1 to 4Y	11	31,7	13	12				
P35	5C1 to 5Y	18	15,36	16	37				
P36	6C1 to 6Y	18	15,36	19	20				
P37	1C2 to 1Y	23	31,5	3	21				
P38	2C2 to 2Y	23	31,5	6	4				
P39	3C2 to 3Y	7	31,11	30	9				
P40	4C2 to 4Y	7	31,11	34	12				
P41	5C2 to 5Y	36	15,18	35	37				
P42	6C2 to 6Y	36	15,18	39	20				
P43	1C3 to 1Y	5,23	31	24	21				
P44	2C3 to 2Y	5,23		27	4				
P45	3C3 to 3Y	7,11		28	9				
P46	4C3 to 4Y	7,11		32	12				
P47	5C3 to 5Y	18,36	15	14	37				
P48	6C3 to 6Y	18,36	15	38	20				
P49	1A-2A to 1Y	2	22,23,31	5	21	10	41		
P50	1A-2A to 2Y	26	23,25,31	5	4				
P51	3A-4A to 3Y	29	7,8,31	11	9				
P52	3A-4A to 4Y	13	7,31,33	11	12				
P53	5A-6A to 5Y	16	15,17,36	18	37				
P54	5A-6A to 6Y	19	15,36,40	18	20				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P55	1A-2A to 1Y Propagation delay time (t_{PHL})	24,23	31,3	5	21	10	41	ns	14,16
P56	1A-2A to 2Y	27,23	31,6	5	4				
P57	3A-4A to 3Y	20,7	31,30	11	9				
P58	3A-4A to 4Y	32,7	31,34	11	12				
P59	5A-6A to 5Y	14,36	15,35	18	37				
P60	5A-6A to 6Y	38,36	15,39	18	20				
P61	1B-2B to 1Y	3	5,22,31	23	21				
P62	1B-2B to 2Y	6	5,25,31	23	4				
P63	3B-4B to 3Y	30	8,11,31	7	9				
P64	3B-4B to 4Y	34	11,31,33	7	12				
P65	5B-6B to 5Y	35	15,17,18	36	37				
P66	5B-6B to 6Y	39	15,18,40	36	20				
P67	1B-2B to 1Y	24,5	31,2	23	21				
P68	1B-2B to 2Y	27,5	31,26	23	4				
P69	3B-4B to 3Y	28,11	31,29	7	9				
P70	3B-4B to 4Y	32,11	31,13	7	12				
P71	5B-6B to 5Y	14,18	15,16	36	37				
P72	5B-6B to 6Y	38,18	15,19	36	20				
P73	1A-2A to 1Y	22	2,23,31	5	21	7			14,17
P74	1A-2A to 2Y	25	23,26,31	5	4				
P75	3A-4A to 3Y	8	7,29,31	11	9				
P76	3A-4A to 4Y	33	7,13,31	11	12				
P77	5A-6A to 5Y	17	15,16,36	18	37				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P78	5A-6A to 6Y Propagation delay time (t_{PHL})	40	15,19,36	18	20	7	41	ns	14,17
P79	1A-2A to 1Y	3,23	24,31	5	21				
P80	1A-2A to 2Y	6,23	27,31	5	4				
P81	3A-4A to 3Y	30,7	28,31	11	9				
P82	3A-4A to 4Y	34,7	32,31	11	12				
P83	5A-6A to 5Y	35,36	14,15	18	37				
P84	5A-6A to 6Y	39,36	38,15	18	20				
P85	1B-2B to 1Y	22	3,5,31	23	21				
P86	1B-2B to 2Y	25	5,6,31	23	4				
P87	3B-4B to 3Y	8	11,30,31	7	9				
P88	3B-4B to 4Y	33	11,31,34	7	12				
P89	5B-6B to 5Y	17	15,18,35	36	37				
P90	5B-6B to 6Y	40	15,18,39	36	20				
P91	1B-2B to 1Y	5,2	24,31	23	21				
P92	1B-2B to 2Y	5,26	27,31	23	4				
P93	3B-4B to 3Y	11,29	28,31	7	9				
P94	3B-4B to 4Y	11,13	32,31	7	12				
P95	5B-6B to 5Y	18,16	14,15	36	37				
P96	5B-6B to 6Y	18,19	38,15	36	20				

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TABLE XVI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1G-4G to 1Y Propagation delay time (t_{PLH})	22	5,23	31	21	10	43	ns	14,17
Q2	1G-4G to 2Y	25	5,23		4				
Q3	1G-4G to 3Y	8	7,11		9				
Q4	1G-4G to 4Y	33	7,11		12				
Q5	5G-6G to 5Y	17	18,36	15	37				
Q6	5G-6G to 6Y	40	18,36	15	20				
Q7	1G-4G to 1Y	2,5	23	31	21				
Q8	1G-4G to 2Y	26,5	23		4				
Q9	1G-4G to 3Y	29,11	7		9				
Q10	1G-4G to 4Y	13,11	7		12				
Q11	5G-6G to 5Y	16,18	36	15	37				
Q12	5G-6G to 6Y	19,18	36	15	20				
Q13	1G-4G to 1Y	3,23	5	31	21				
Q14	1G-4G to 2Y	6,23	5		4				
Q15	1G-4G to 3Y	30,7	11		9				
Q16	1G-4G to 4Y	34,7	11		12				
Q17	5G-6G to 5Y	35,36	18	15	37				
Q18	5G-6G to 6Y	39,36	18	15	20				
Q19	1G-4G to 1Y	5,23,24	Open	31	21				
Q20	1G-4G to 2Y	5,23,27			4				
Q21	1G-4G to 3Y	7,11,28			9				
Q22	1G-4G to 4Y	7,11,32			12				
Q23	5G-6G to 5Y	18,14,36		15	37				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q24	5G-6G to 6Y Propagation delay time (t_{PLH})	18,36,38	Open	15	20	10	43	ns	14,17
Q25	1C0 to 1Y	Open	5,23,31	22	21	5	29		14,16
Q26	2C0 to 2Y		5,23,31	25	4				
Q27	3C0 to 3Y		7,11,31	8	9				
Q28	4C0 to 4Y		7,11,31	33	12				
Q29	5C0 to 5Y		18,15,36	17	37				
Q30	6C0 to 6Y		18,15,36	40	20				
Q31	1C1 to 1Y	5	23,31	2	21				
Q32	2C1 to 2Y	5	23,31	26	4				
Q33	3C1 to 3Y	11	7,31	29	9				
Q34	4C1 to 4Y	11	7,31	13	12				
Q35	5C1 to 5Y	18	15,36	16	37				
Q36	6C1 to 6Y	18	15,36	19	20				
Q37	1C2 to 1Y	23	5,31	3	21				
Q38	2C2 to 2Y	23	5,31	6	4				
Q39	3C2 to 3Y	7	11,31	30	9				
Q40	4C2 to 4Y	7	11,31	34	12				
Q41	5C2 to 5Y	36	15,18	35	37				
Q42	6C2 to 6Y	36	15,18	39	20				
Q43	1C3 to 1Y	5,23	31	24	21				
Q44	2C3 to 2Y	5,23		27	4				
Q45	3C3 to 3Y	7,11		28	9				
Q46	4C3 to 4Y	7,11		32	12				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q47	5C3 to 5Y Propagation delay time (t_{PLH})	18,36	15	14	37	5	29	ns	14,16
Q48	6C3 to 6Y	18,36	15	38	20	5	29		
Q49	1A-2A to 1Y	2	22,23,31	5	21	10	41		
Q50	1A-2A to 2Y	26	23,25,31	5	4				
Q51	3A-4A to 3Y	29	7,8,31	11	9				
Q52	3A-4A to 4Y	13	7,31,33	11	12				
Q53	5A-6A to 5Y	16	15,17,36	18	37				
Q54	5A-6A to 6Y	19	15,36,40	18	20				
Q55	1A-2A to 1Y	23,24	3,31	5	21				
Q56	1A-2A to 2Y	23,27	6,31	5	4				
Q57	3A-4A to 3Y	7,28	30,31	11	9				
Q58	3A-4A to 4Y	7,32	31,34	11	12				
Q59	5A-6A to 5Y	14,36	15,35	18	37				
Q60	5A-6A to 6Y	36,38	15,39	18	20				
Q61	1B-2B to 1Y	3	5,22,31	23	21				
Q62	1B-2B to 2Y	6	5,25,31	23	4				
Q63	3B-4B to 3Y	30	8,11,31	7	9				
Q64	3B-4B to 4Y	34	11,31,33	7	12				
Q65	5B-6B to 5Y	35	15,17,18	36	37				
Q66	5B-6B to 6Y	39	15,18,40	36	20				
Q67	1B-2B to 1Y	5,24	2,31	23	21				
Q68	1B-2B to 2Y	5,27	26,31	23	4				
Q69	3B-4B to 3Y	11,28	29,31	7	9				

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TABLE I . Propagation delay time (t_{PLH}) test - Continued.

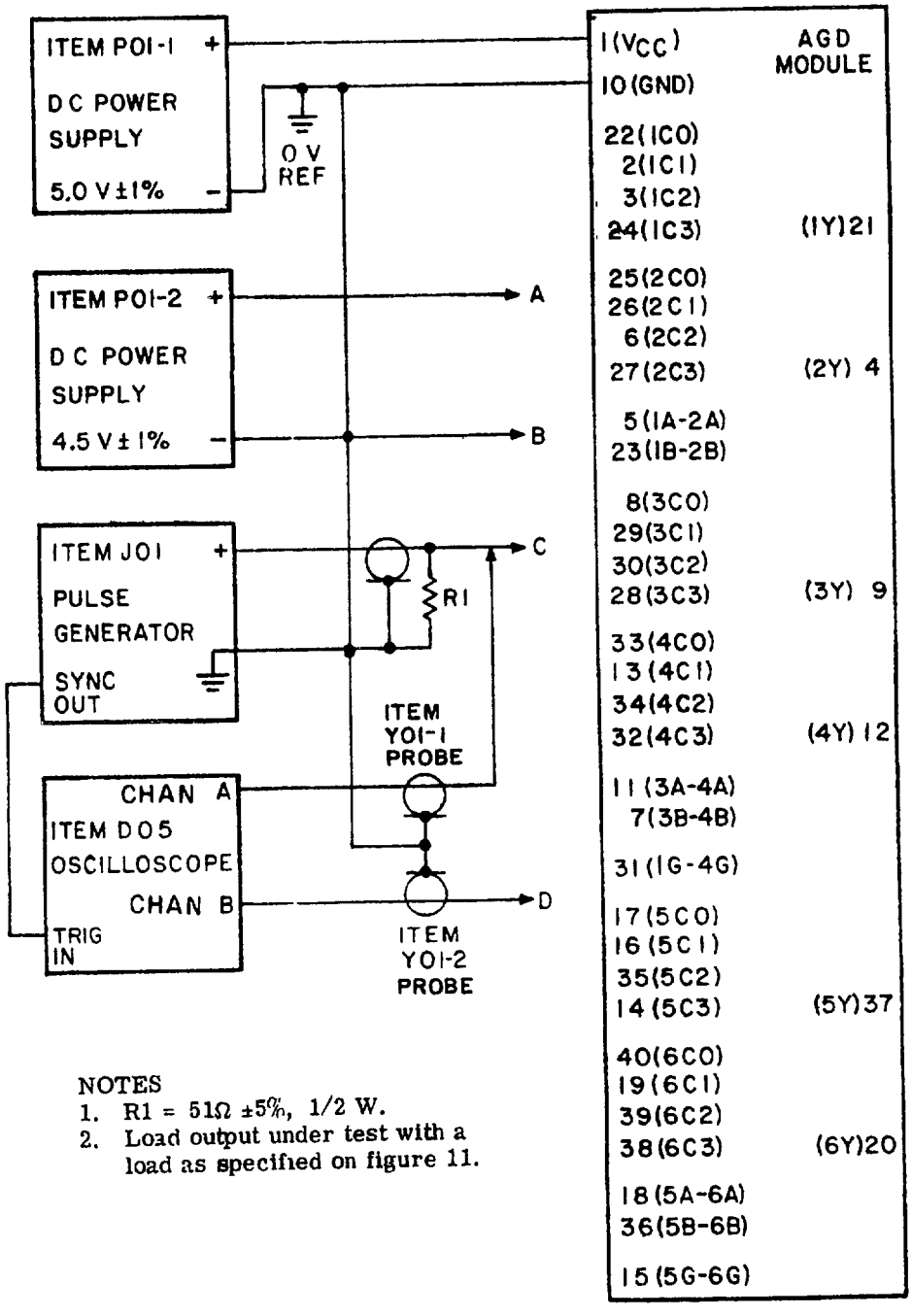
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
070	3B-4B to 4Y Propagation delay time (t_{PLH})	11,22	10,31	7	12	10	41	ns	14,16
071	5B-6B to 5Y	14,18	15,16	36	37		41		14,16
072	5B-6B to 6Y	18,28	15,19	36	20		41		14,16
073	1A-2A to 1Y	22	2,23,31	5	21		39		14,17
074	1A-2A to 2Y	25	23,26,31	5	4				
075	3A-4A to 3Y	8	7,29,31	11	9				
076	3A-4A to 4Y	33	7,13,31	11	12				
077	5A-6A to 5Y	17	15,16,36	18	37				
078	5A-6A to 6Y	40	15,19,36	18	20				
079	1A-2A to 1Y	8,23	20,31	5	21				
080	1A-2A to 2Y	6,23	27,31	5	4				
081	3A-4A to 3Y	7,30	29,31	11	9				
082	3A-4A to 4Y	7,30	31,32	11	12				
083	5A-6A to 5Y	25,36	14,15	18	37				
084	5A-6A to 6Y	29,36	15,38	18	20				
085	1B-2B to 1Y	22	3,5,31	23	21				
086	1B-2B to 2Y	25	5,6,31	23	4				
087	3B-4B to 3Y	8	11,30,31	7	9				
088	3B-4B to 4Y	33	11,31,32	7	12				
089	5B-6B to 5Y	17	15,16,36	36	37				
090	5B-6B to 6Y	40	15,19,36	36	20				
091	1B-2B to 1Y	8,23	20,31	23	21				
092	1B-2B to 2Y	6,23	27,31	23	4				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q93	3B-4B to 3Y Propagation delay time (t_{PLH})	11,29	28,31	7	9	10	39	ns	14,17
Q94	3B-4B to 4Y	11,13	31,32	7	12				
Q95	5B-6B to 5Y	16,18	14,15	36	37				
Q96	5B-6B to 6Y	18,19	15,38	36	20				

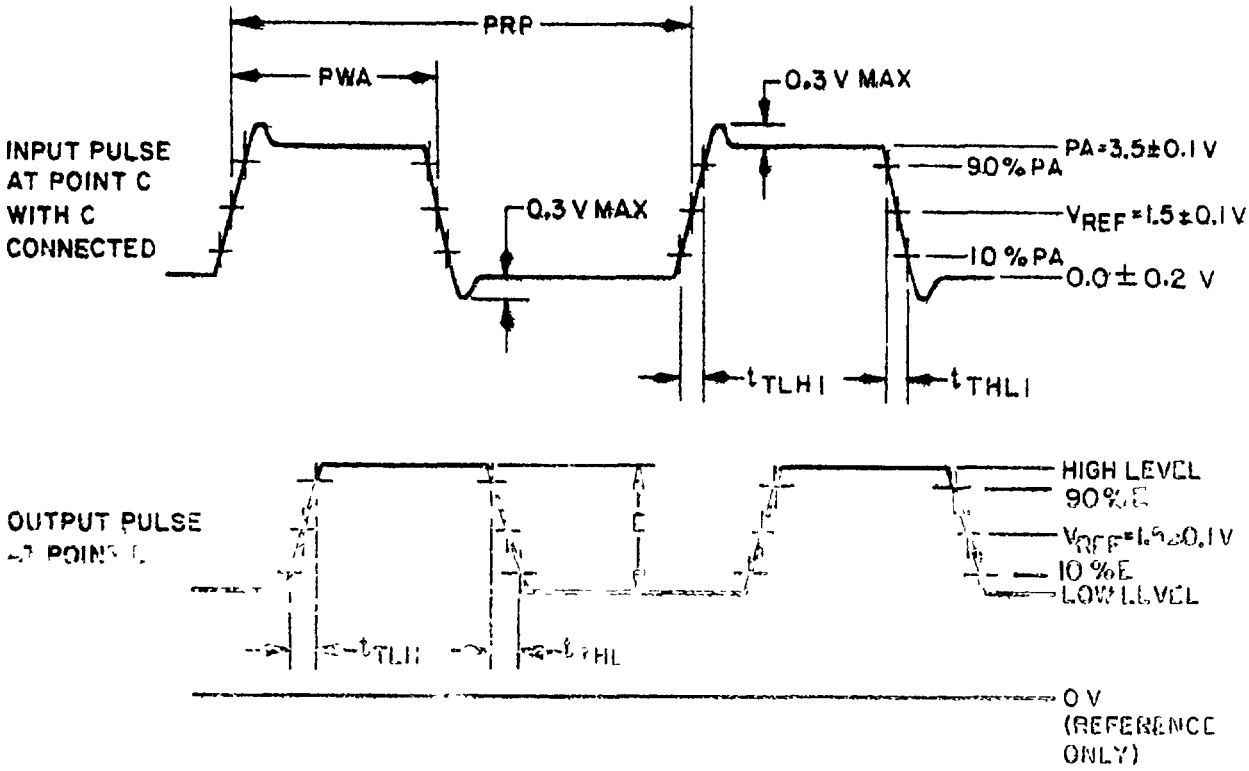
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NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. Load output under test with a load as specified on figure 11.

* FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XIII, XIV, XV, and XVI.

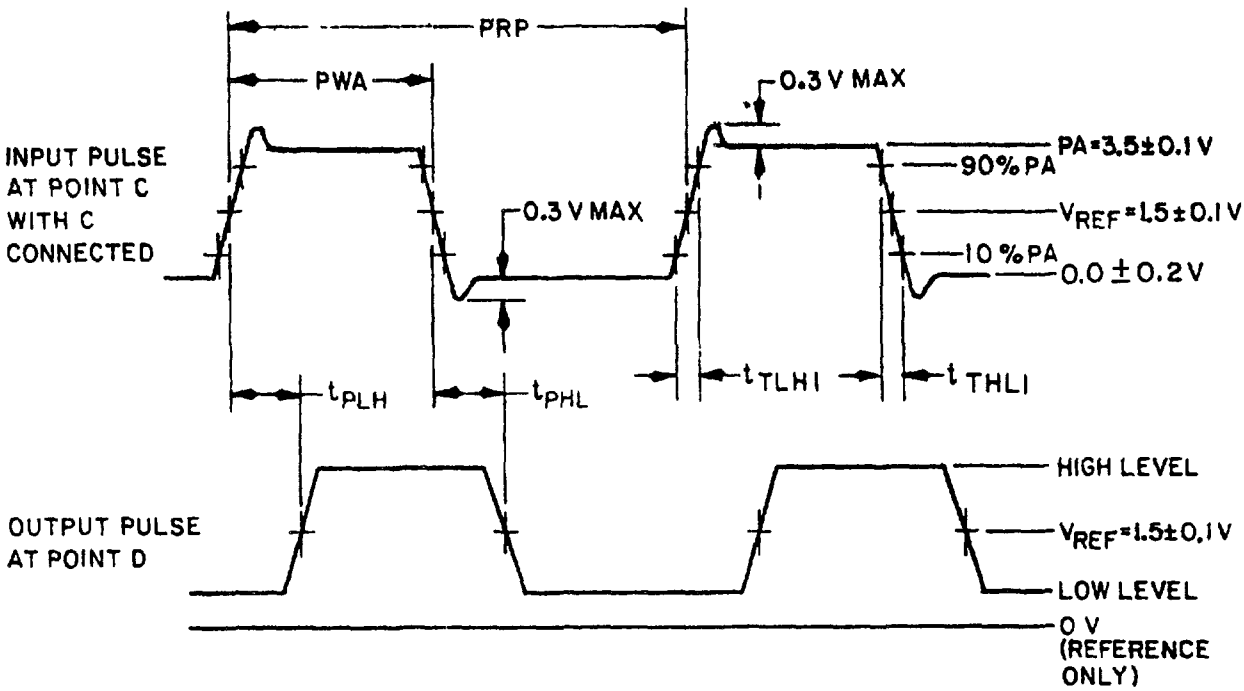
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 200 ±20 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XIII and XIV.

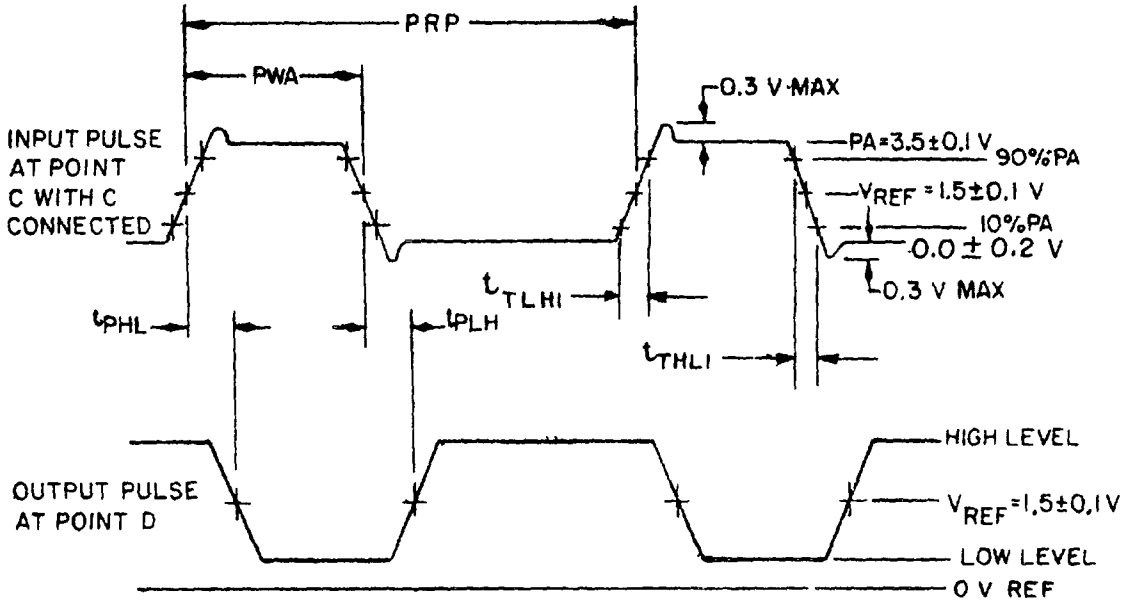
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NOTES
1. Input pulse parameters:
   $t_{TLHI} = t_{THLI} = 10 \pm 2$ ns.
   PWA = 200 ±20 ns.
   PRP = 1,000 ±100 ns
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.

* FIGURE 16. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XV and XVI.

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NOTES.
1. Input pulse parameters.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 200 ±20 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.

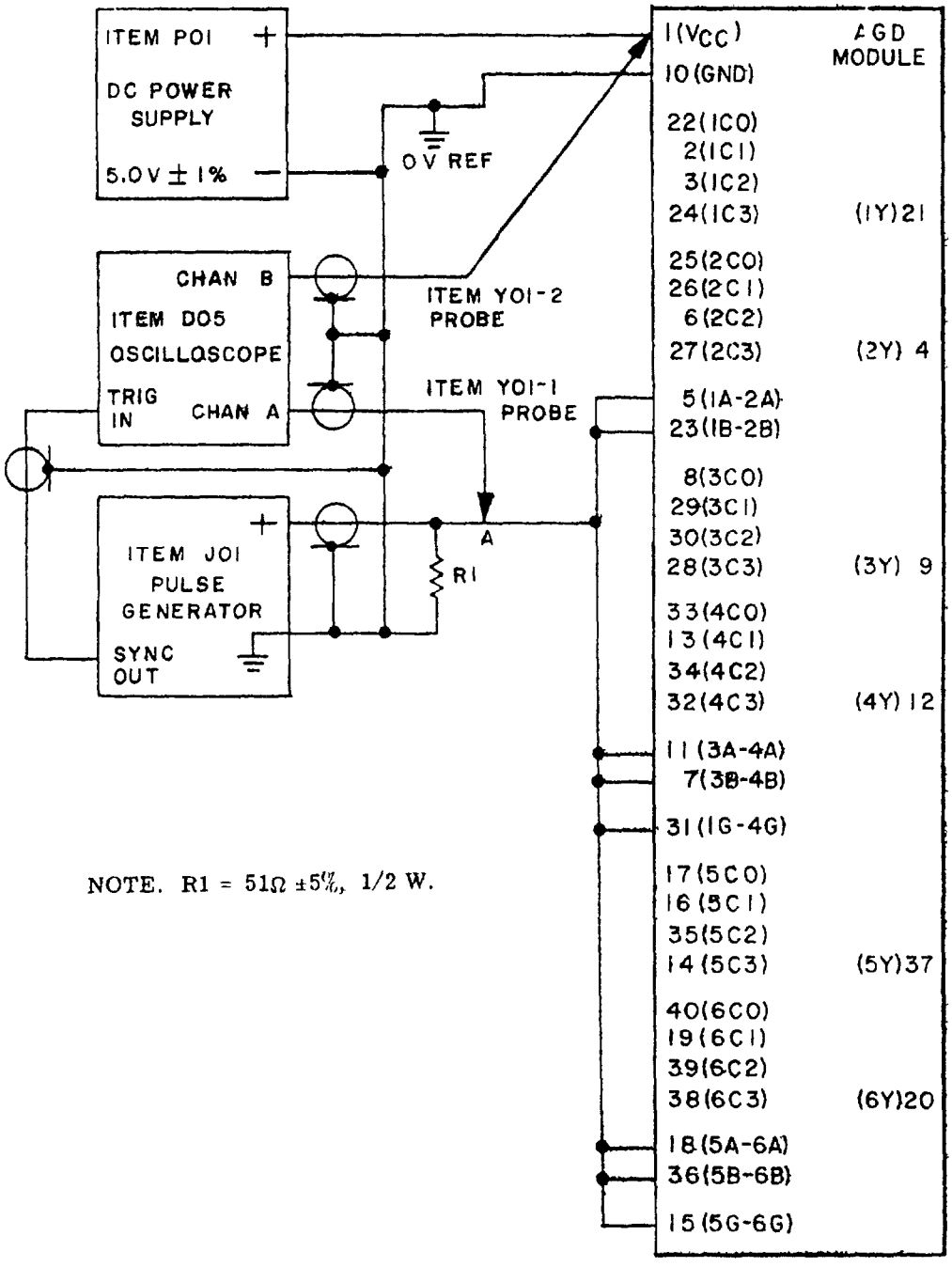
* FIGURE 17. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XV and XVI.

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TABLE XVII. Power su - transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply transient amplitude	0	400	mV	18, 19

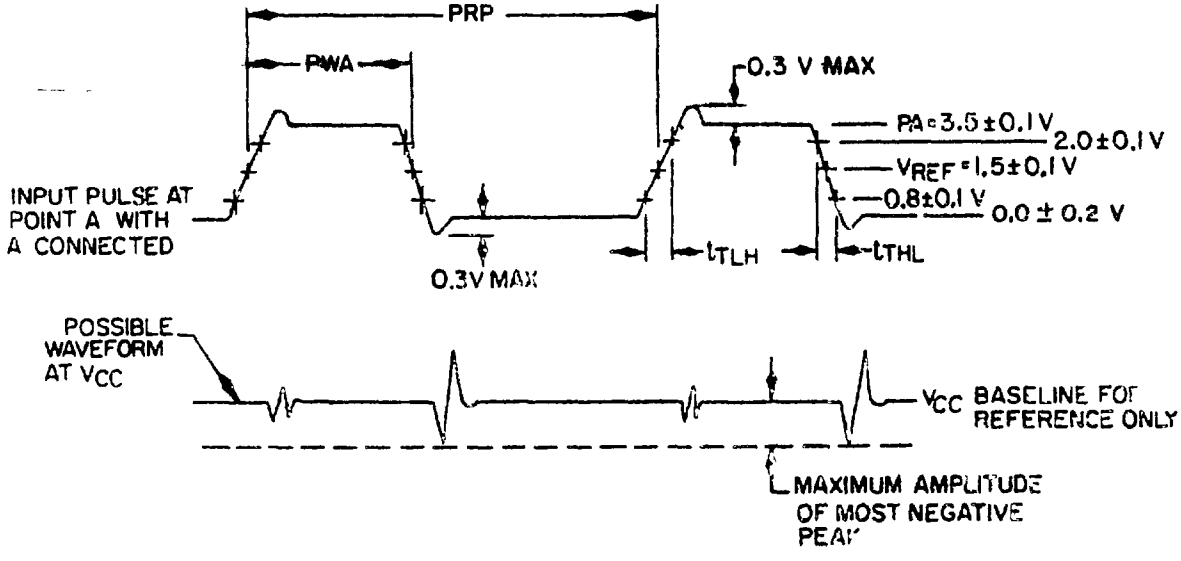
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NOTE. R1 = 51Ω ±5%, 1/2 W.

* FIGURE 18 Power supply transient amplitude test circuit for table XVII.

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NOTE. Input pulse parameters:
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
PWA = 200 ±20 ns.
PRP = 1,000 ±100 ns.

* FIGURE 19. Power supply transient amplitude test circuit waveforms for table XVII.

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TABLE XVIII. Output crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1Y Output crosstalk, low level	8,17,25,33, 40	22,5,7,11, 15,18,23, 31,36	21	Open	0	500	mV	20,21
S2	2Y	8,17,22, 33,40	25,5,7,11, 15,18,23, 31,36	4					
S3	3Y	17,22,25, 33,40	8,5,7,11, 15,18,23, 31,36	9					
S4	4Y	8,17,22,25, 40	33,5,7,11, 15,18,23, 31,36	12					
S5	5Y	8,22,25,33, 40	17,5,7,11, 15,18,23, 31,36	37					
S6	6Y	8,17,22,25, 33	40,5,7,11, 15,18,23, 31,36	20					

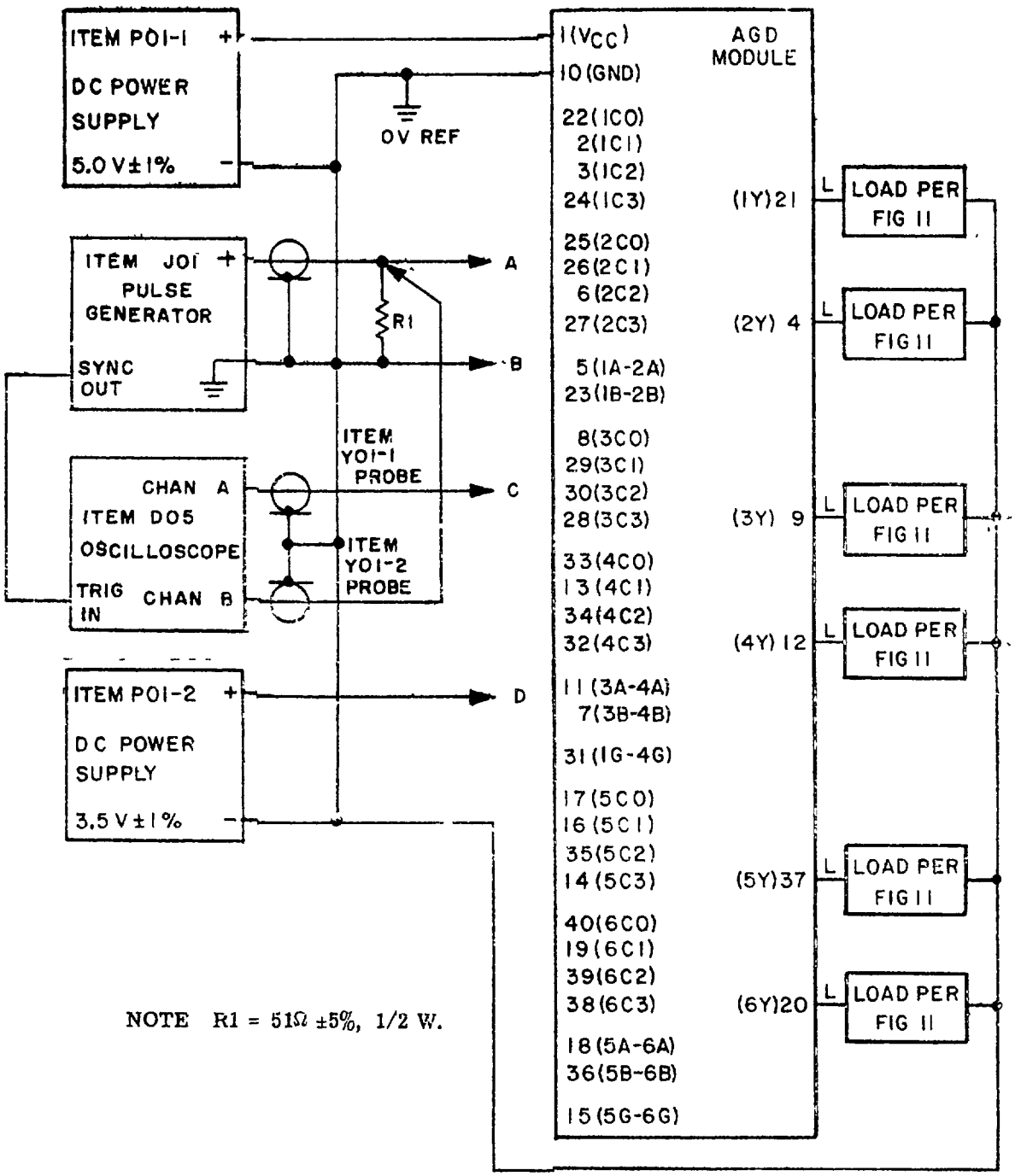
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TABLE XIX. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1Y Output crosstalk, high level	8,17,25,33, 40	5,7,11,15, 18,23,31, 36	21	22	0	400	mV	20,21
T2	2Y	8,17,22,33, 40	5,7,11,15, 18,23,31, 36	4	25				
T3	3Y	17,22,25, 33,40	5,7,11,15, 18,23,31, 36	9	8				
T4	4Y	8,17,22,25, 40	5,7,11,15, 18,23,31, 36	12	33				
T5	5Y	8,22,25,33, 40	5,7,11,15, 18,23,31, 36	37	17				
T6	6Y	8,17,22,25, 33	5,7,11,15, 18,23,31, 36	20	40				

5/

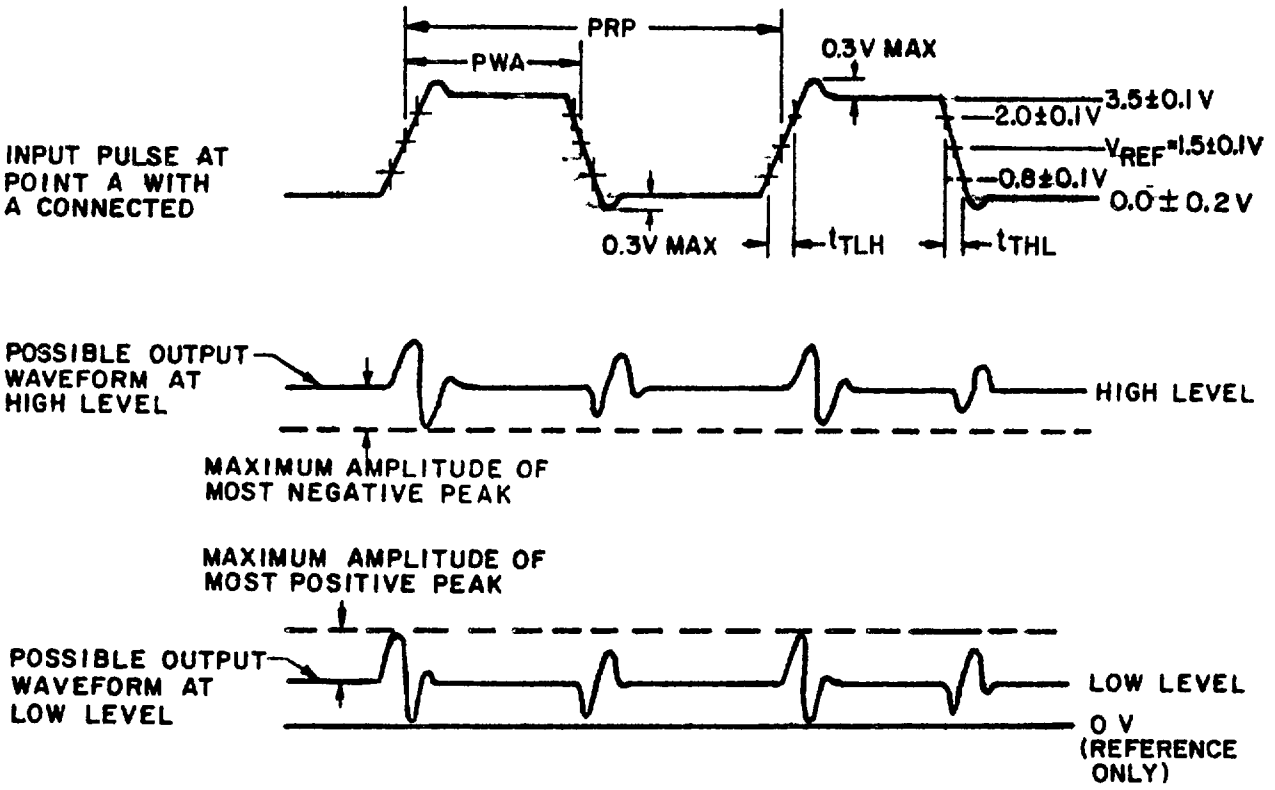
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NOTE R1 = 51Ω ±5%, 1/2 W.

* FIGURE 20. Output crosstalk test circuit for tables XVIII and XIX

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 200 ±20 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of probes Y01-1 and Y01-2 and test fixture and module ground pins, shall be less than 1 inch (25.4 mm).

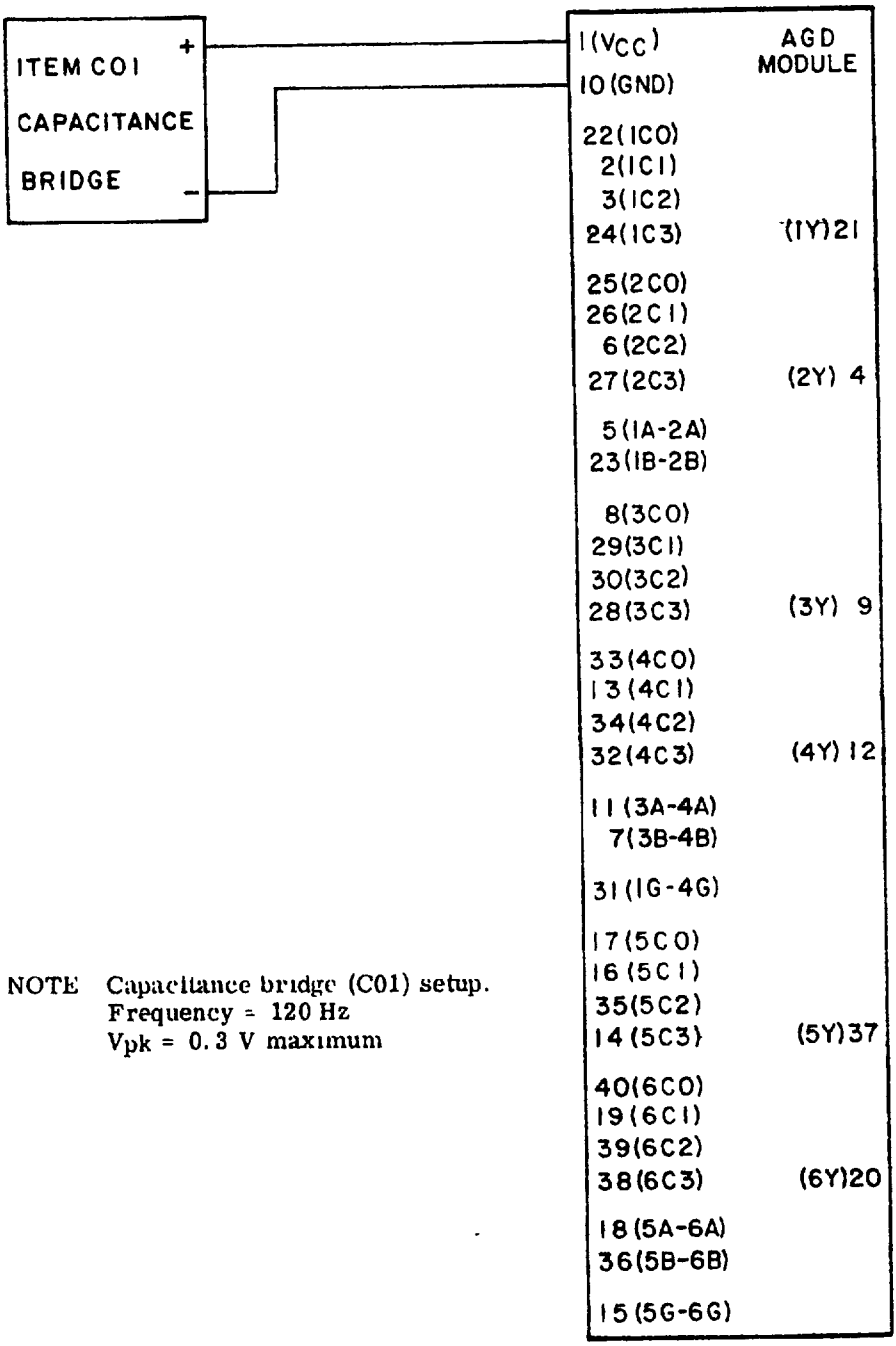
* FIGURE 21. Output crosstalk test circuit waveforms for tables XVIII and XIX.

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TABLE XX. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply filter capacitance	2.2	6.6	µF	22

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NOTE Capacitance bridge (C01) setup.
Frequency = 120 Hz
Vpk = 0.3 V maximum

FIGURE 22. Power supply filter capacitance test circuit for table XX.

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*TABLE XXI. Functional test.

Test figure	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
V1	Functional	2,6,8,13,16,17,19,22, 26,27,29,32,34,38,39	3,5,7,11,14,15,18,23, 24,25,28,30,31,33,35, 36,40	2/, 3/	H	L	H	L	H	L	← 1/ →			23
V2		2,3,5,6,8,11,13,14,16, 17,18,19,22,24,25,26, 27,28,29,30,32,33,34, 35,38,39,40	7,15,23,31,36		H	H	H	H	H	H				
V3		2,7,8,14,16,22,23,24, 25,27,28,29,32,33,36, 38,40	3,5,6,11,13,15,17,18, 19,26,30,31,34,35,39		L	L	L	L	L	L				
V4		3,6,7,13,14,19,23,24, 26,28,30,34,35,36,39	2,5,8,11,15,16,17,18, 22,25,27,29,31,32,33, 38,40		H	H	H	H	H	H				
V5		2,8,14,15,16,17,22,24, 25,27,28,29,31,32,33, 38,40	3,5,6,7,11,13,18,19, 23,26,30,34,35,36,39		L	L	L	L	L	L				
V6		2,3,5,6,11,13,15,16, 18,19,25,26,29,30,31, 33,34,35,39,40	7,8,14,17,22,23,24, 27,28,32,36,38		L	L	L	L	L	L				
V7		2,3,16,25,29,30,33,35, 40	5,6,7,8,11,13,14,15, 17,18,19,22,23,24,26, 27,28,31,32,34,36,38, 39		L	H	L	H	L	H				
V8		3,5,6,7,8,11,13,14,17, 18,19,22,23,24,26,28, 30,33,34,35,36,39	2,15,16,25,27,29,31, 32,38,40		H	L	H	L	H	L				
V9		3,5,6,8,11,14,17,18,22, 24,25,27,28,30,32,33, 34,35,38,39,40	2,7,13,15,16,19,23, 26,29,31,36		L	L	L	L	L	L				
V10	↓	2,3,6,7,15,16,23,25,27, 29,30,31,32,33,34,35, 36,38,39,40	5,8,11,13,14,17,18, 19,22,24,26,28	↓	L	L	L	L	L	L	↓			↓

See footnotes at end of table.

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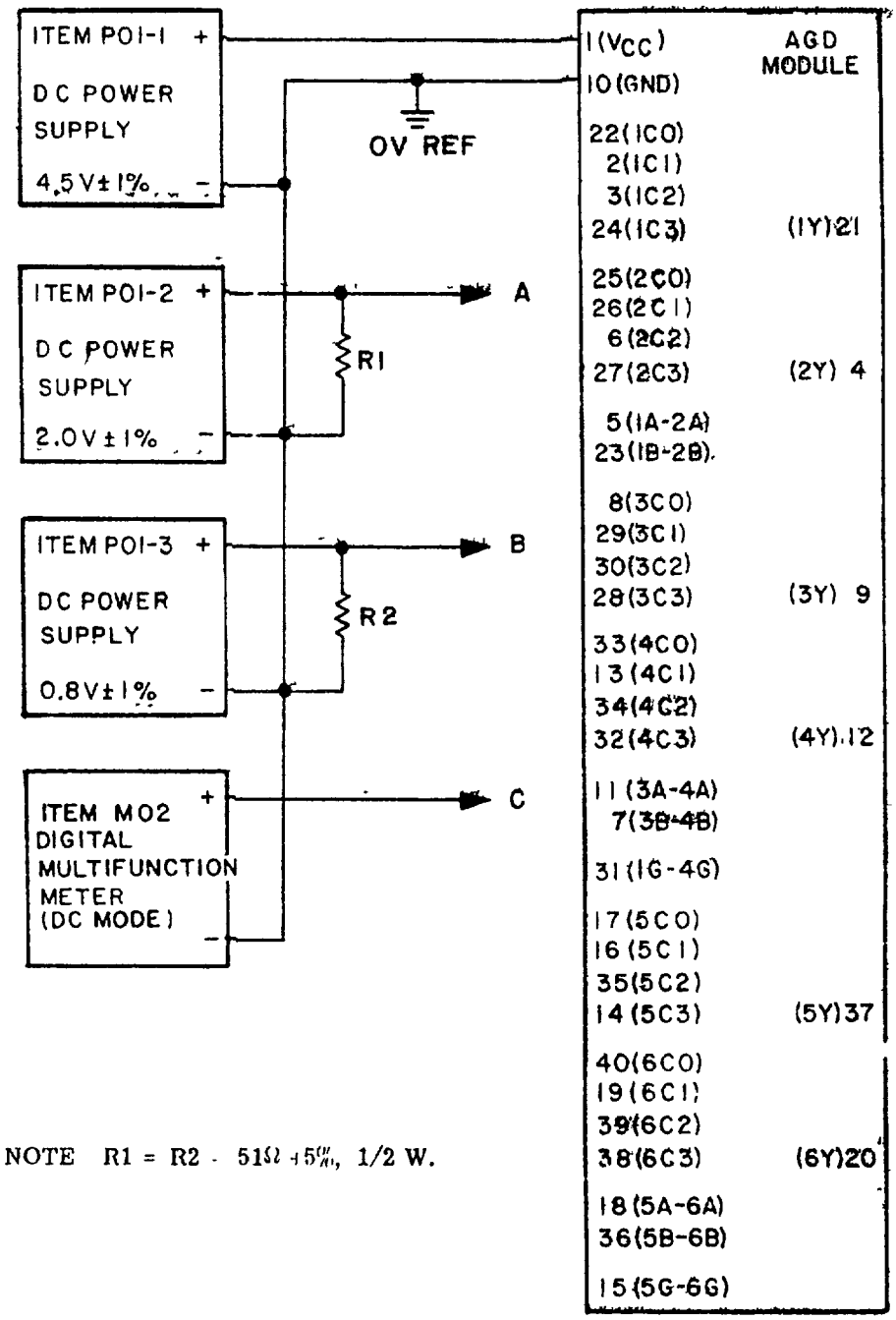
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*TABLE XXI. Functional test - Continued.

Test figure	Parameter	Connect			Output logic states						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Y	2Y	3Y	4Y	5Y	6Y	Min	Max		
V11	Functional	2,3,5,6,7,11,16,18,23, 25,27,29,30,32,33,34, 35,36,38,39,40	8,13,14,15,17,19,22, 24,26,28,31	2/, 3/	L	H	L	H	L	H	← 1/ →			23
V12	Functional	2,3,5,6,7,8,11,13,14, 15,16,17,18,19,22,23, 24,26,27,28,29,30,31, 32,34,35,36,38,39	25,33,40	2/, 3/	L	L	L	L	L	L	← 1/ →			23
V13	Functional	24,26,6,27,8,29,30,28, 11,7,33,13,34,17,16,36	22,2,3,5,23,31,25, 32,35,10,18,15,40, 19,39,38	2/, 3/	L	L	H	L	L	L	← 1/ →			23

1/ Output logic states: L = Low level = 0.0 V to 0.4 V, H = High level = 2.4 V to 4.5 V.
2/ For test conditions V1-V13, verification of the output logic states in footnote 1 is obtained by connecting item M02, point C, to each output pin in succession after input conditions have been accomplished.
3/ 1Y, 2Y, 3Y, 4Y, 5Y, 6Y output logic states are measured at output pins 21, 4, 9, 12, 37, and 20, respectively.

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NOTE R1 = R2 = 51Ω ±5%, 1/2 W.

* FIGURE 23 Functional test circuit for table XXI

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